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English ▼

VPC Fundamentals

Amazon Virtual Private Cloud (Amazon VPC) lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including selection of your own IP address range, creation of subnets, and configuration of route tables and network gateways. You can use both IPv4 and IPv6 in your VPC for secure and easy access to resources and applications.

In this lab, you will learn how to set up a VPC with public and private subnets, connect it to the internet and establish private connections to AWS services.

 **Warning**
Please complete the pre-requisites section before starting this lab.

 If you are running this lab at an **AWS-led event**, your facilitator has already set the region for you. If you are running it **self-service**, simply use the region you are working in. In either case, the region shown in the screenshots may not match your environment — this will not cause any problems.

If you are running this lab in **your own AWS Account**, it is recommended for all lab resources to be created in us-east-1 region so that the screenshots match your environment. This is not mandatory.

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▶ Connecting to SDWAN

▶ Network Monitoring

▶ Network analytics using Amazon Quick Suite

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Contributors & Revision History

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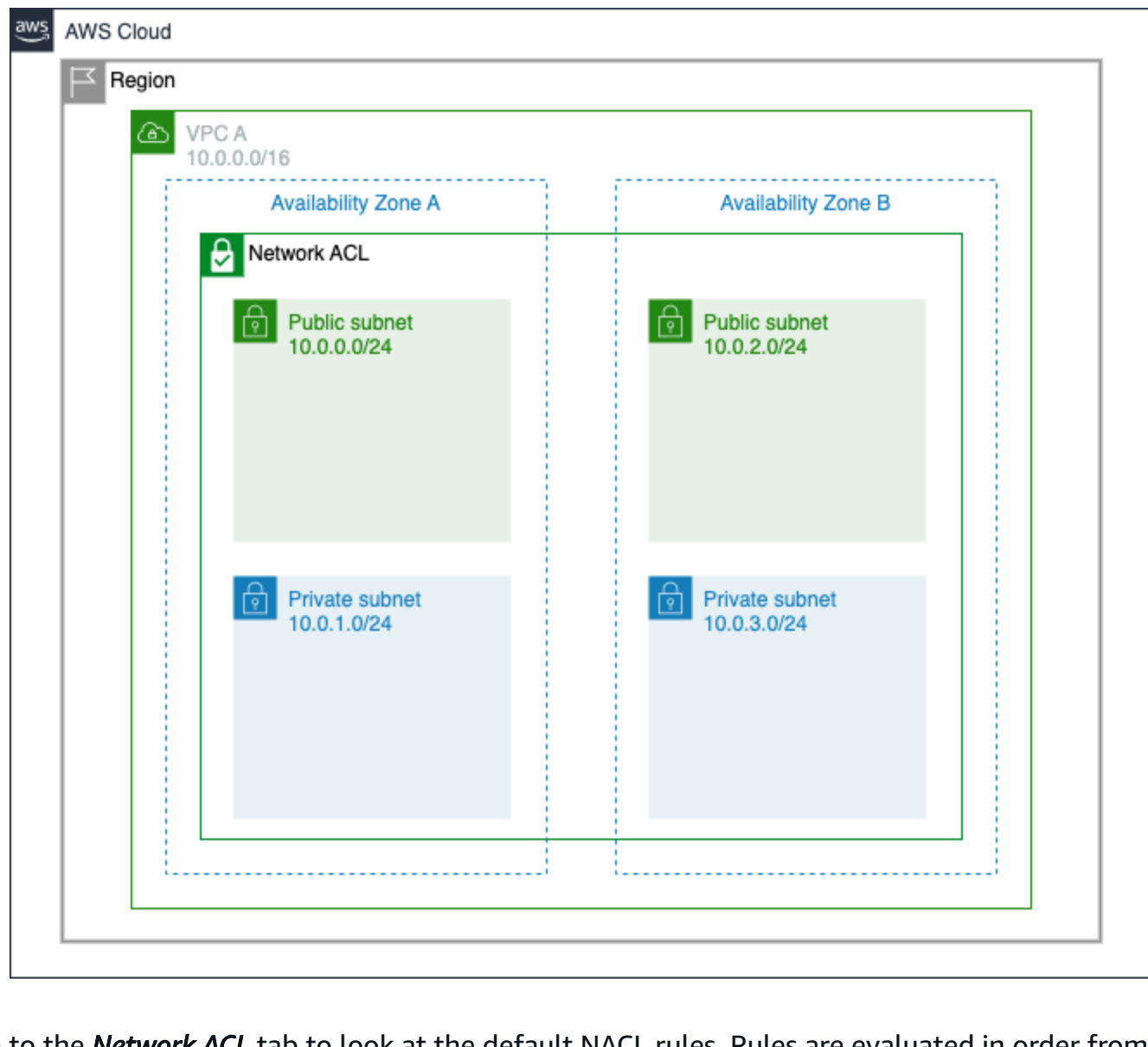
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Language

English

Network ACLs

A network access control list (ACL) is an optional layer of security for your VPC for controlling traffic in and out of one or more subnets.



Select any of the subnets, and scroll down to the **Network ACL** tab to look at the default NACL rules. Rules are evaluated in order from lowest to highest. If the traffic doesn't match any rules, the * rule is applied, and the traffic is denied. Default NACLs allow all inbound and outbound traffic, as shown below, unless customized.

Subnets (1/4) Info

Name	Subnet ID	State	VPC	IPv4 CIDR	IPv6 CIDR
VPC A Public Subnet...	subnet-0...	Available	vpc-	10.0.0.0/24	-
VPC A Private Subnet...	subnet-0...	Available	vpc-	10.0.1.0/24	-
VPC A Public Subnet...	subnet-0...	Available	vpc-	10.0.2.0/24	-
VPC A Private Subnet...	subnet-0...	Available	vpc-	10.0.3.0/24	-

subnet-01b6c3824cab25e8 / VPC A Public Subnet AZ1

DetailsFlow logsRoute tableNetwork ACLCIDR reservationsSharingTags

Network ACL: acl-0

Edit network ACL association

Inbound rules (2)

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Filter inbound rules

Outbound rules (2)

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Filter outbound rules

Create a new Network ACL for workload subnets in VPC A

- On the VPC Dashboard click on [Network ACLs](#)
- Click **Create network ACL**

Network ACLs (1) Info

Create network ACL

Name	Network ACL ID	Associated w...	Default	VPC ID	Inbound rules c...	Outbound rules count
-	acl-0	4 Subnets	Yes	vpc-	2 Inbound rules	2 Outbound rules

- In the **Network ACL** settings screen

- Enter a name of VPC A workload Subnets NACL
- Select VPC A from the dropdown
- Click **Create network ACL**

Create network ACL Info

A network ACL is an optional layer of security that acts as a firewall for controlling traffic in and out of a subnet.

Network ACL settings

Name - optional

Creates a tag with a key of 'Name' and a value that you specify.

VPC A Workloads Subnet NACL

VPC

VPC to use for this network ACL

vpc- (VPC A)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Add new tag

You can add 49 more tags.

CancelCreate network ACL

The result will be a new NACL for VPC A alongside the default NACL created when the VPC was created.

- In the resulting **Network ACLs** screen

- Select the checkbox for VPC A workload Subnets NACL
- Scroll down to the **Subnet associations** tab
- Click **Edit subnet associations**

You successfully created acl-0 / VPC A Workloads Subnet NACL.

Network ACLs (1/2) Info

Create network ACL

Name	Network ACL ID	Associated with	Default	VPC ID
VPC A Workloads Subnet NACL	acl-0	-	No	vpc- / VPC A
-	acl-0	4 Subnets	Yes	vpc- / VPC A

acl-0321d1a316be0cfc / VPC A Workloads Subnet NACL

DetailsInbound rulesOutbound rulesSubnet associationsTags

Subnet associations

Edit subnet associations

Name	Subnet ID	Associated with	Availability Zone	IPv4 CIDR	IPv6 CIDR
No subnets in this region are associated with this network ACL.					

- In the **Edit subnet associations** screen

- Select all four VPC A subnets to associate them with the NACL
- Click **Save changes**.

Edit subnet associations Info

Change which subnets are associated with this network ACL.

Available subnets (4/4)

Filter subnet associations

Name	Subnet ID	Associated with	Availability Zone	IPv4 CIDR
VPC A Public Subnet AZ1	subnet-0	acl-0	us-east-1a	10.0.0.0/24
VPC A Private Subnet AZ1	subnet-0	acl-0	us-east-1a	10.0.1.0/24
VPC A Public Subnet AZ2	subnet-0	acl-0	us-east-1b	10.0.2.0/24
VPC A Private Subnet AZ2	subnet-0	acl-0	us-east-1b	10.0.3.0/24

Selected subnets

subnet-0 / VPC A Public Subnet AZ1

subnet-0 / VPC A Public Subnet AZ2

subnet-0 / VPC A Private Subnet AZ1

subnet-0 / VPC A Private Subnet AZ2

CancelSave changes

The NACL should now be associated with four subnets on the following screen but because NACLs are created with only a DENY rule for inbound and outbound we will now change the default NACL rules to allow all traffic in both directions.

- In the **Network ACLs** screen

- Select the check box for VPC A workload Subnets NACL for VPC A
- Scroll down and select the **Inbound Rules** tab below
- Notice that we have only DENY all rule
- Click **Edit inbound rules**

Network ACLs (1/2) Info

Create network ACL

Name	Network ACL ID	Associated with	Default	VPC ID
VPC A Workloads Subnet NACL	acl-0	4 Subnets	No	vpc- / VPC A
-	acl-0	-	Yes	vpc- / VPC A

acl-0 / VPC A Workloads Subnet NACL

DetailsInbound rulesOutbound rulesSubnet associationsTags

Inbound rules (1)

Edit inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

Filter inbound rules

- In **Edit inbound rules** screen

- Click **Add new rule**
- Input **100** in **Rule number**
- Choose **All traffic** in **Type**
- Leave **Source** as **0.0.0.0/0**
- Click **Save changes**

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Filter inbound rules

Add new ruleSort by rule number

CancelPreview changesSave changes

- In the resulting screen you should have a success banner and a new Allow rule under the **Inbound rules** tab:

You have successfully updated inbound rules for acl-0 / VPC A Workloads Subnet NACL.

Network ACLs (1/2) Info

Create network ACL

Name	Network ACL ID	Associated with	Default	VPC ID
VPC A Workloads S...	acl-0	4 Subnets	No	vpc- / VPC A
-	acl-0	-	Yes	vpc- / VPC A

acl-0 / VPC A Workloads Subnet NACL

DetailsInbound rulesOutbound rulesSubnet associationsTags

Inbound rules (2)

Edit inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Filter inbound rules

Now follow the same steps described above for Inbound, but work on **Outbound Rules** tab of NACLs

- On the **Outbound Rules** tab

- Note that we have only DENY all rule
- Click **Edit outbound rules**

Network ACLs (1/2) Info

Create network ACL

Name	Network ACL ID	Associated with	Default	VPC ID
VPC A Workloads S...	acl-0	4 Subnets	No	vpc- / VPC A
-	acl-0	-	Yes	vpc- / VPC A

acl-0 / VPC A Workloads Subnet NACL

DetailsInbound rulesOutbound rulesSubnet associationsTags

Outbound rules (1)

Edit outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

Filter outbound rules

- In the **Edit outbound rules** screen

- Click **Add new rule**
- Input **100** in **Rule number**
- Choose **All traffic** in **Type**
- Leave **Destination** as **0.0.0.0/0**
- Click **Save changes**

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the VPC.

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Filter outbound rules

Add new ruleSort by rule number

CancelPreview changesSave changes

- On the resulting screen check that the rule has been added under the **Outbound rules** tab

You have successfully updated outbound rules for acl-0 / VPC A Workloads Subnet NACL.

Network ACLs (1/2) Info

Create network ACL

Name	Network ACL ID	Associated with	Default	VPC ID
VPC A Workloads S...	acl-0	4 Subnets	No	vpc- / VPC A
-	acl-0	-	Yes	vpc- / VPC A

acl-0 / VPC A Workloads Subnet NACL

DetailsInbound rulesOutbound rulesSubnet associationsTags

Outbound rules (2)

Edit outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Filter outbound rules

- Allowing all traffic in and out of your subnets is not a good security posture. You can use NACLs to set broad rules and/or DENY rules, and then use **Security Groups** to create fine grained rules. For example, you can deny traffic from specific IPs with NACLs but not with Security Groups.

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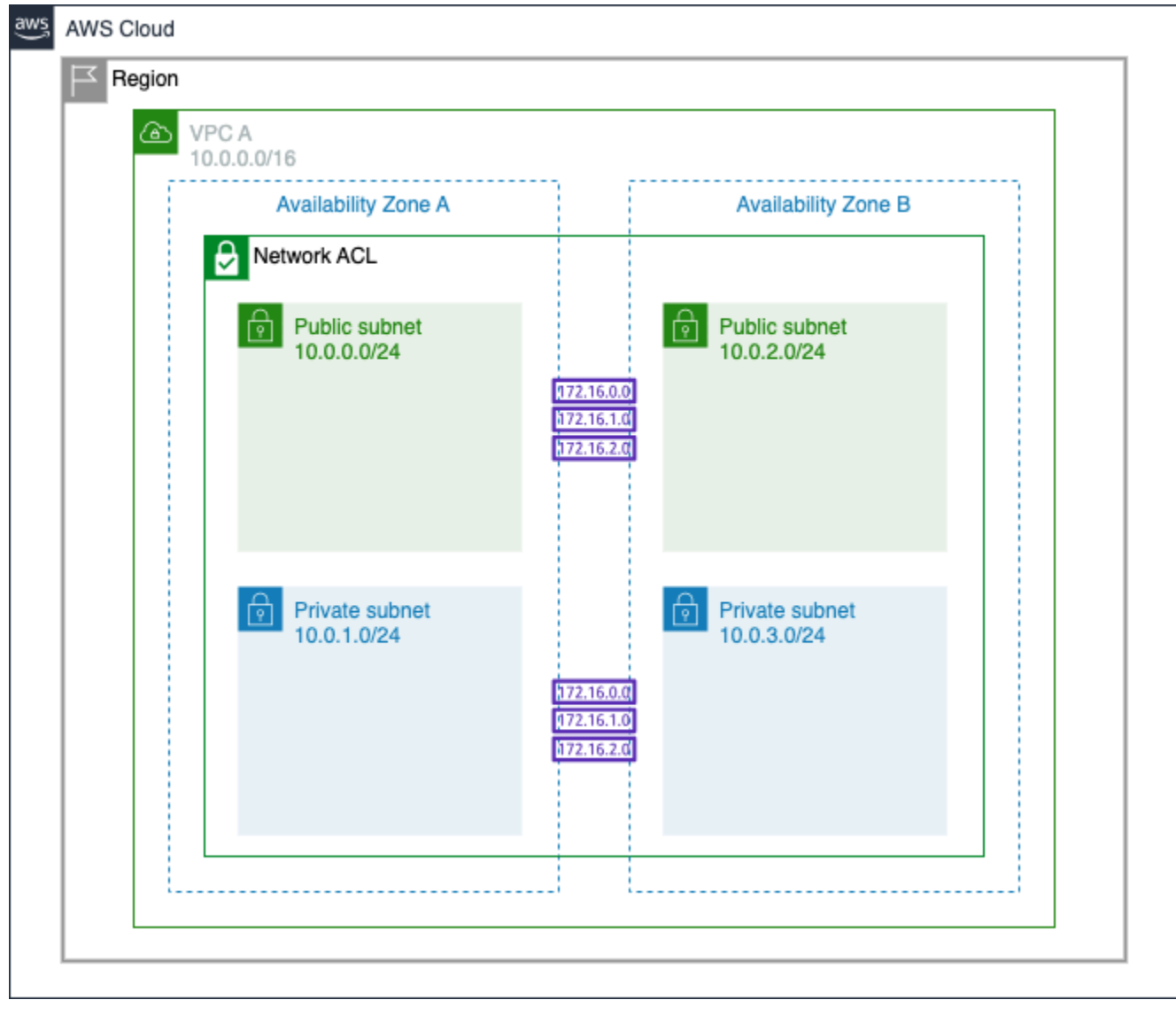
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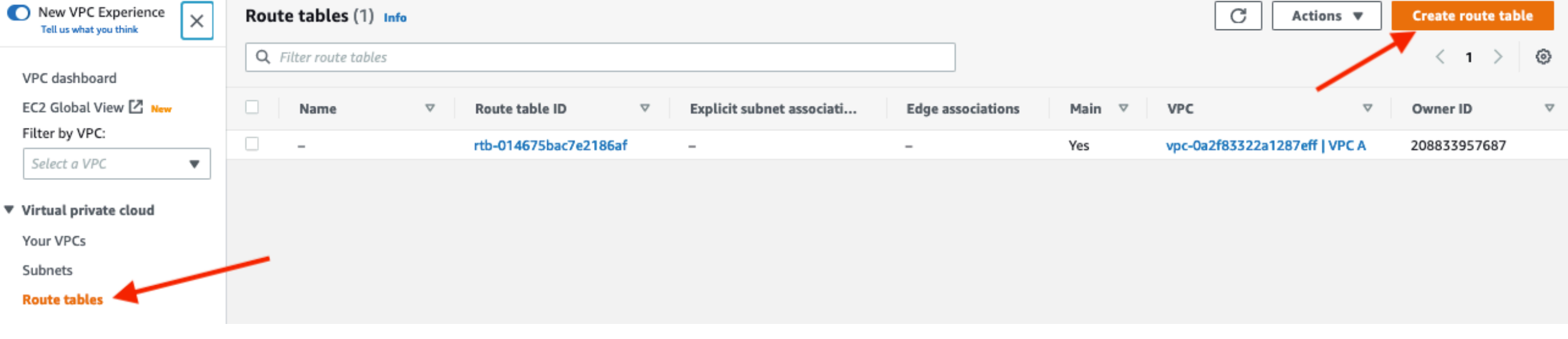
Route Tables

Your VPC has an implicit router, and you use route tables to control where network traffic is directed. Each subnet in your VPC must be associated with a route table, which controls the routing for the subnet (subnet route table). You can explicitly associate a subnet with a particular route table. Otherwise, the subnet is implicitly associated with the main route table. A subnet can only be associated with one route table at a time, but you can associate multiple subnets with the same subnet route table.



Create Route Table for Public Subnets

1. In left hand panel of the VPC Dashboard click on [Route Tables](#)



You will see the default route table that was created as part of the VPC creation, and in the **Subnet Associations** tab below the four subnets created earlier. We will now create a new public route table for the public subnets with a route to the internet via the Internet Gateway.

2. Add new public route table by clicking on **Create route table** in the right hand corner
3. Enter **VPC A Public Route Table** as the name and select **vpc-a** from the VPC dropdown

[VPC](#) > [Route tables](#) > Create route table

Create route table info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a text with a key of 'Name' and a value that you specify.

VPC Public Route Table

VPC
The VPC to use for this route table.

vpc-0: (VPC A)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

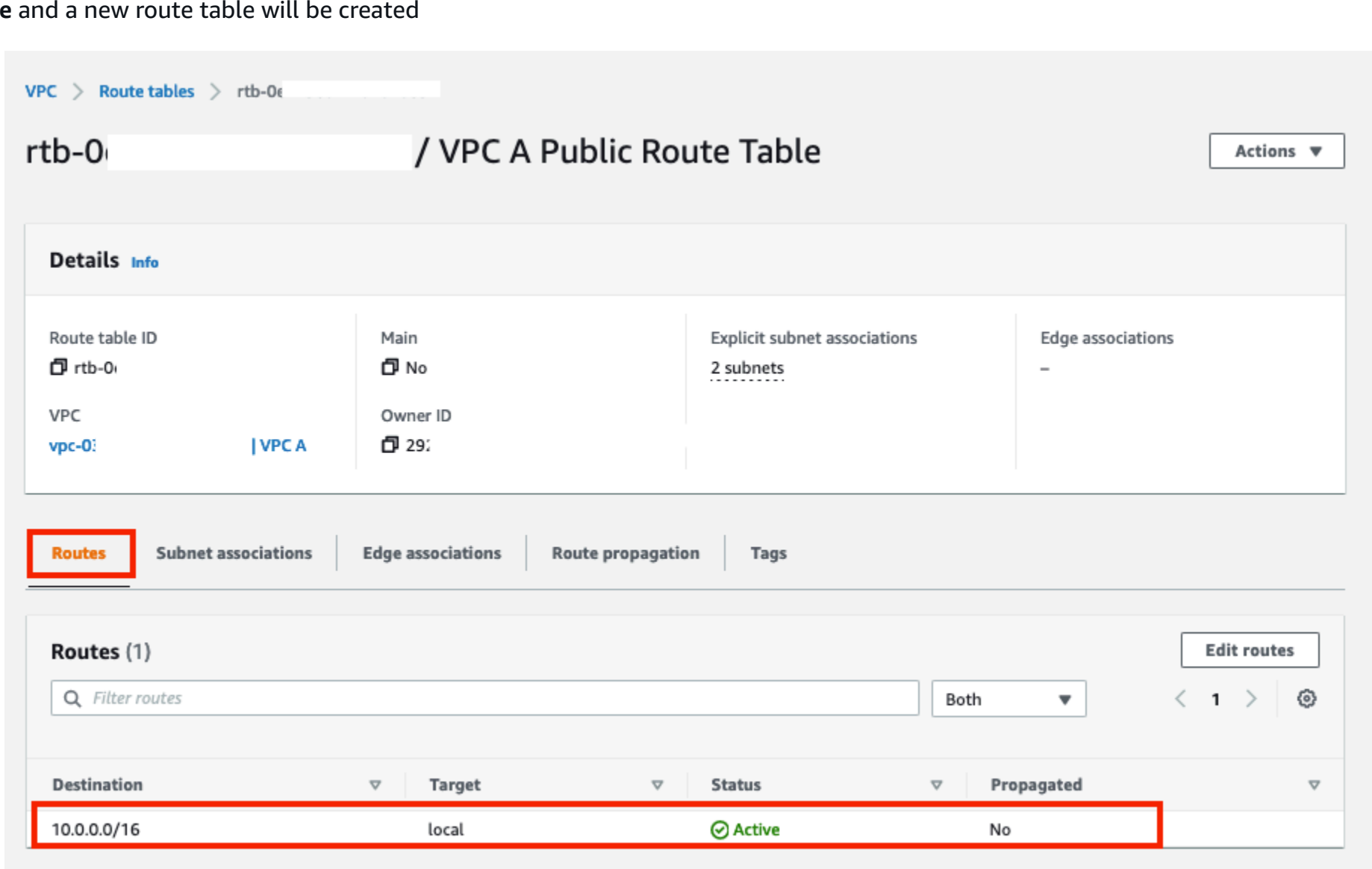
Key **Value - optional!**

Key	Value - optional!
Cl. Name	Cl. VPC A Public Route Table

Add new tag

You can add 48 more tags.

Cancel **Create route table**



As you can see there is only a local route, so we're going to enable internet access by adding a route to an Internet Gateway in a later step. For now we need to associate this public route table with the public subnets we created earlier.

5. Scroll down and click on the **Subnet Associations** tab



6. Click on **Edit subnet associations**

Subnet VPC A Public Subnet A21 and VPC A Public Subnet A22 and click Save association

VPC > Route tables > rtb-01 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

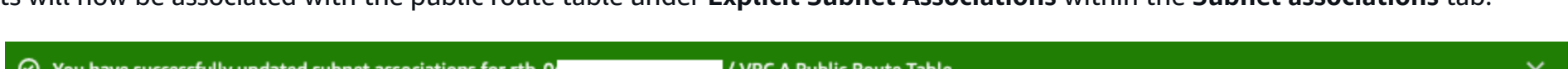
☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID	
☐	VPC A Private Subnet A22	subnet-03	10.0.1.0/24	-	Main (rtb-01)	/ VPC A Main Route Table
☐	VPC A Private Subnet A21	subnet-06	10.0.1.0/24	-	Main (rtb-01)	/ VPC A Main Route Table
☒	VPC A Public Subnet A22	subnet-02	10.0.2.0/24	-	Main (rtb-01)	/ VPC A Main Route Table
☒	VPC A Public Subnet A21	subnet-01	10.0.0.0/24	-	Main (rtb-01)	/ VPC A Main Route Table

Selected subnets

subnet-0 / VPC A Public Subnet A21 X

subnet-0 / VPC A Public Subnet A22 X

Cancel Save associations



9. In the left hand panel of the VPC Dashboard click on [Route Tables](#) and click on the **Create route table** button in the top right corner.

Create route table [info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of "Name" and a value that you specify.

VPC A Private Route Table

VPC

This VPC to use for this route table.

vpc-0 (VPC A)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key Value - optional

Q Name X Q VPC A Private Route Table X Remove

Add new tag

You can add 49 more tags.



- We're going to enable outbound internet access by adding a route to

- No subnet associations

the two private subnets VPC A Private Subnet AZ1 and VPC A Private Subnet AZ2 and click **Save associations**

Edit subnet associations

Change which subnets are associated with this route table.

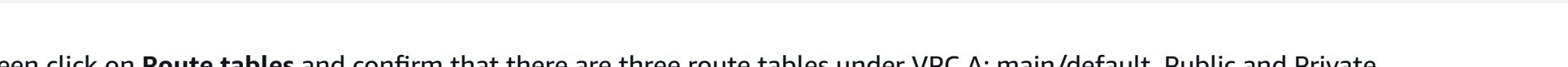
Available subnets (2/4)

Filter subnet associations

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	VPC A Private Subnet AZ2	subnet-01	10.0.5.0/24	–	Main (rtb-c) 0
☒	VPC A Private Subnet AZ1	subnet-00	10.0.1.0/24	–	Main (rtb-c) 0
☐	VPC A Public Subnet AZ2	subnet-0f	10.0.2.0/24	–	rtb-0e / VPC A Public Route Table
☐	VPC A Public Subnet AZ1	subnet-01	10.0.0.0/24	–	rtb-0e / VPC A Public Route Table

Selected subnets

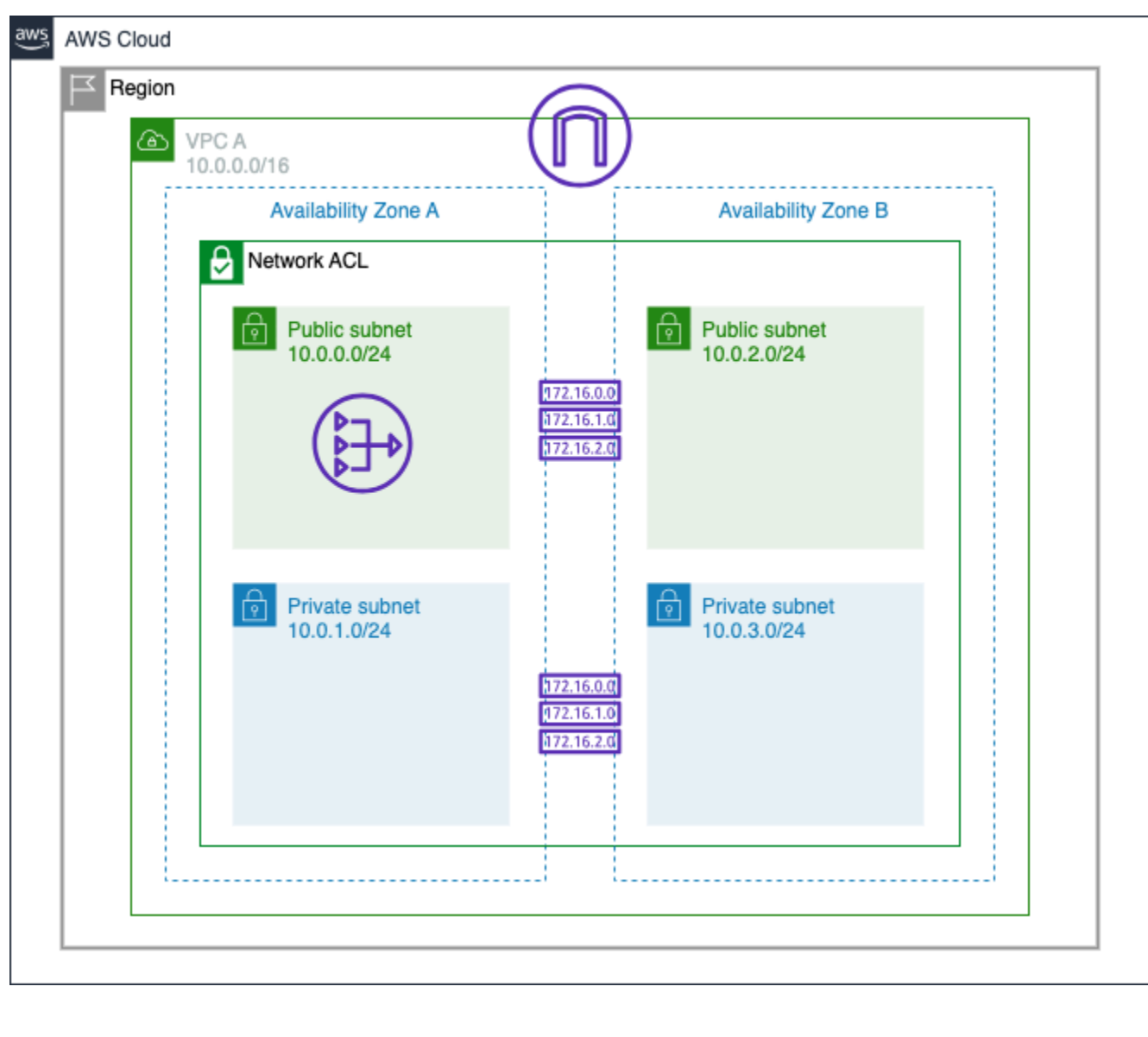
subnet-0 / VPC A Private Subnet AZ1 X subnet-0 / VPC A Private Subnet AZ2 X



Internet Connectivity

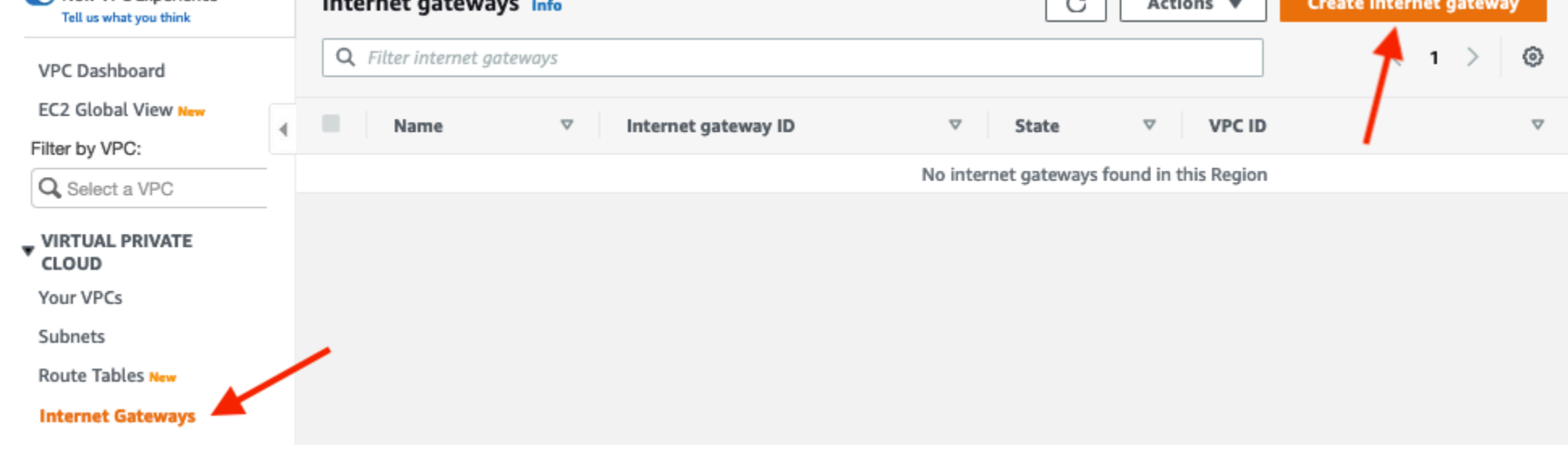
In this section, we will deploy an Internet Gateway (IGW) and NAT Gateway into our VPC.

An Internet Gateway establishes outside connectivity for EC2 instances that will be deployed into the VPC and provides both inbound and outbound connectivity to workloads running in public subnets whereas a NAT Gateway provides outbound connectivity for workloads running in private subnets.

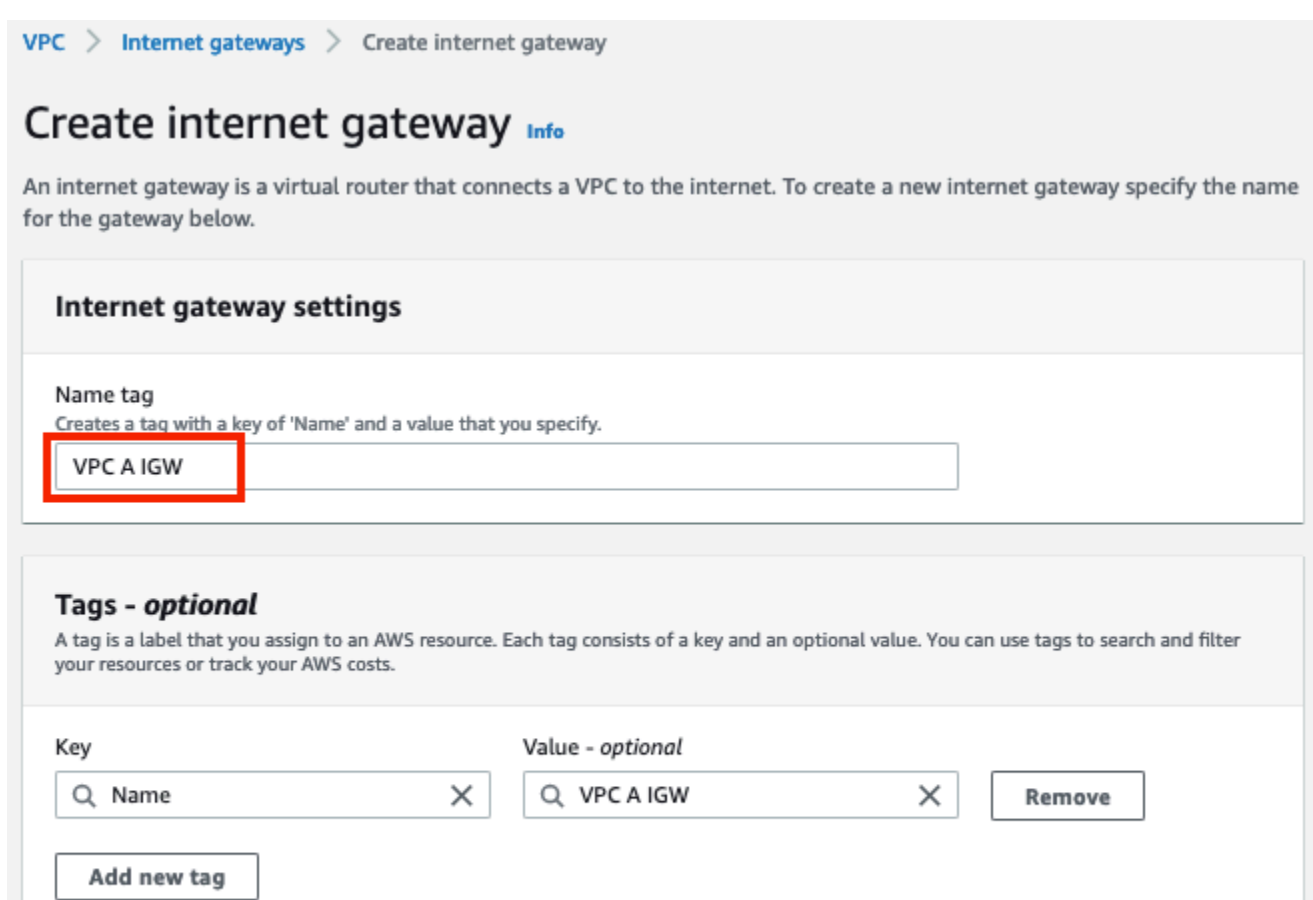


Deploy an Internet Gateway

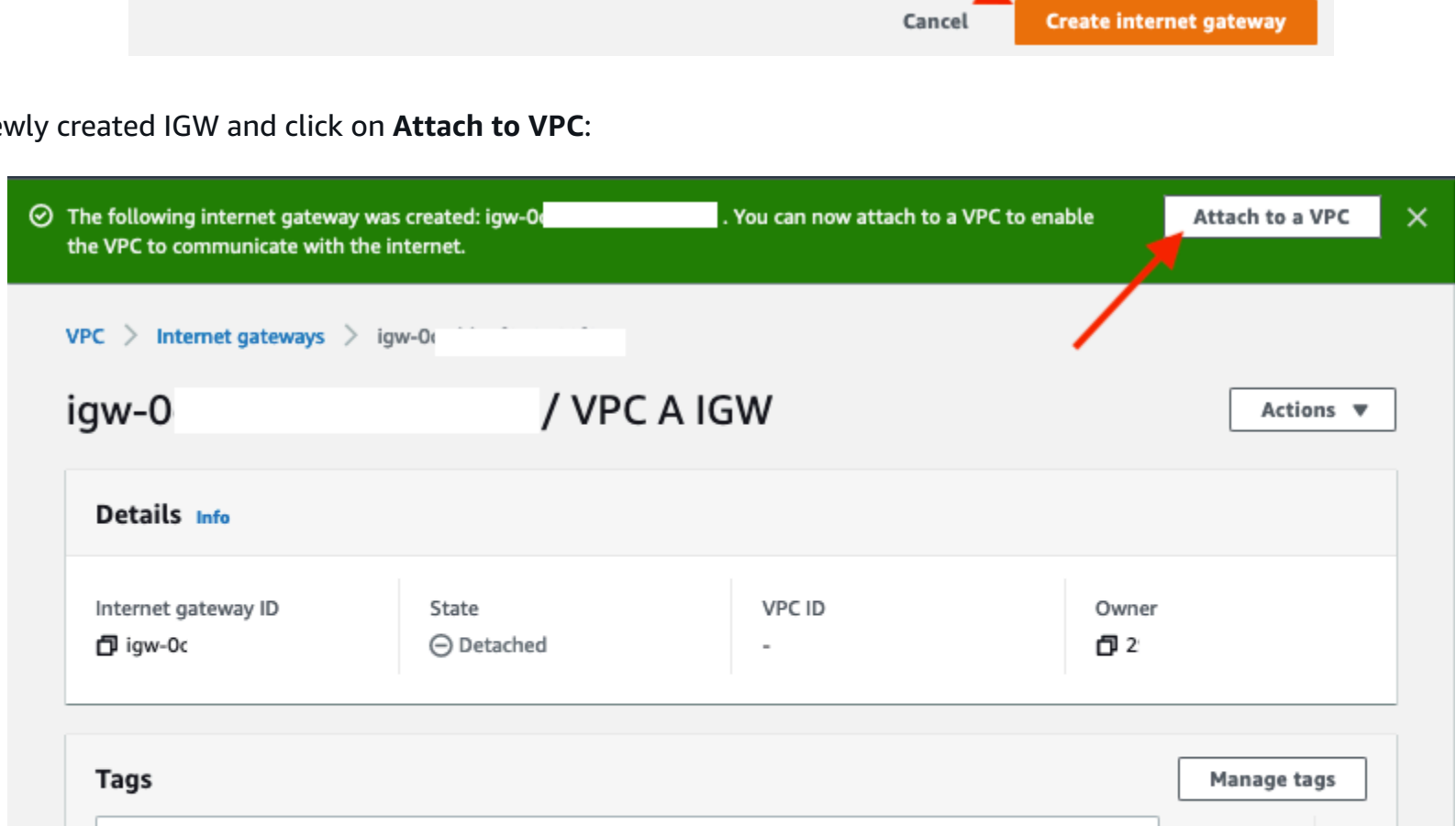
- In the left hand panel click on [Internet Gateways](#) and click on [Create internet gateway](#)



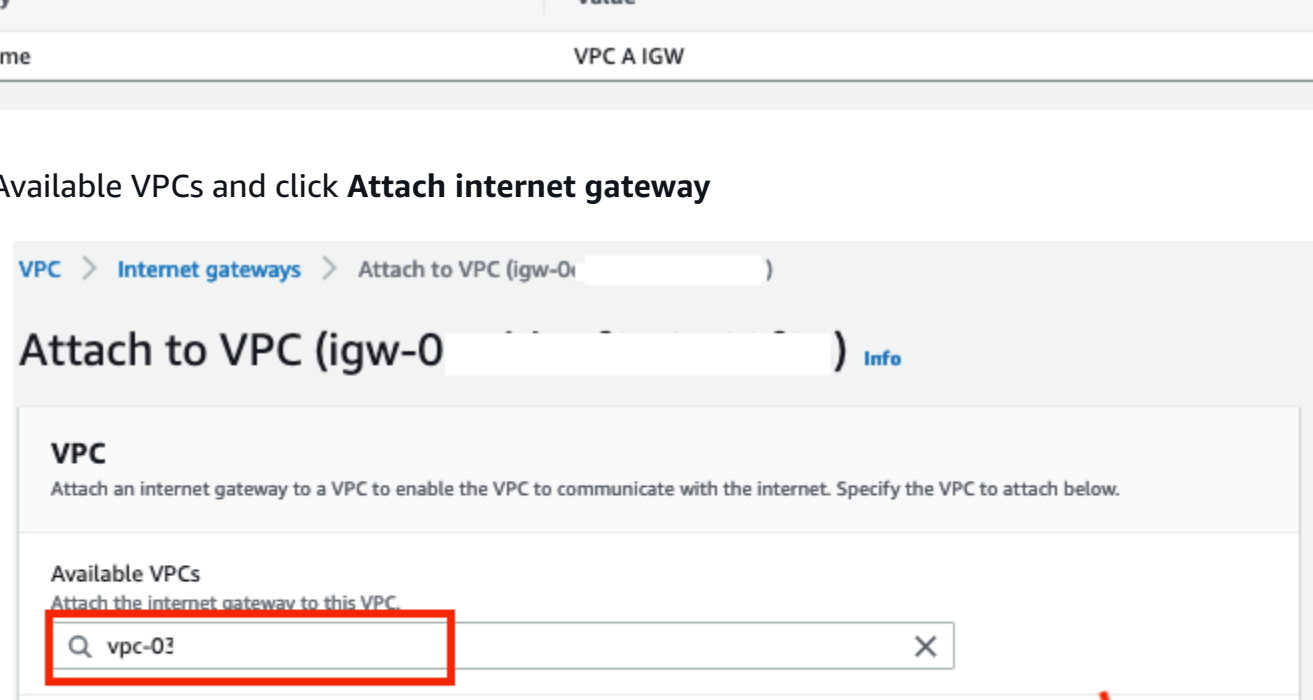
- Enter vpc-a-1gw as the name and click [Create internet gateway](#) in the bottom right corner



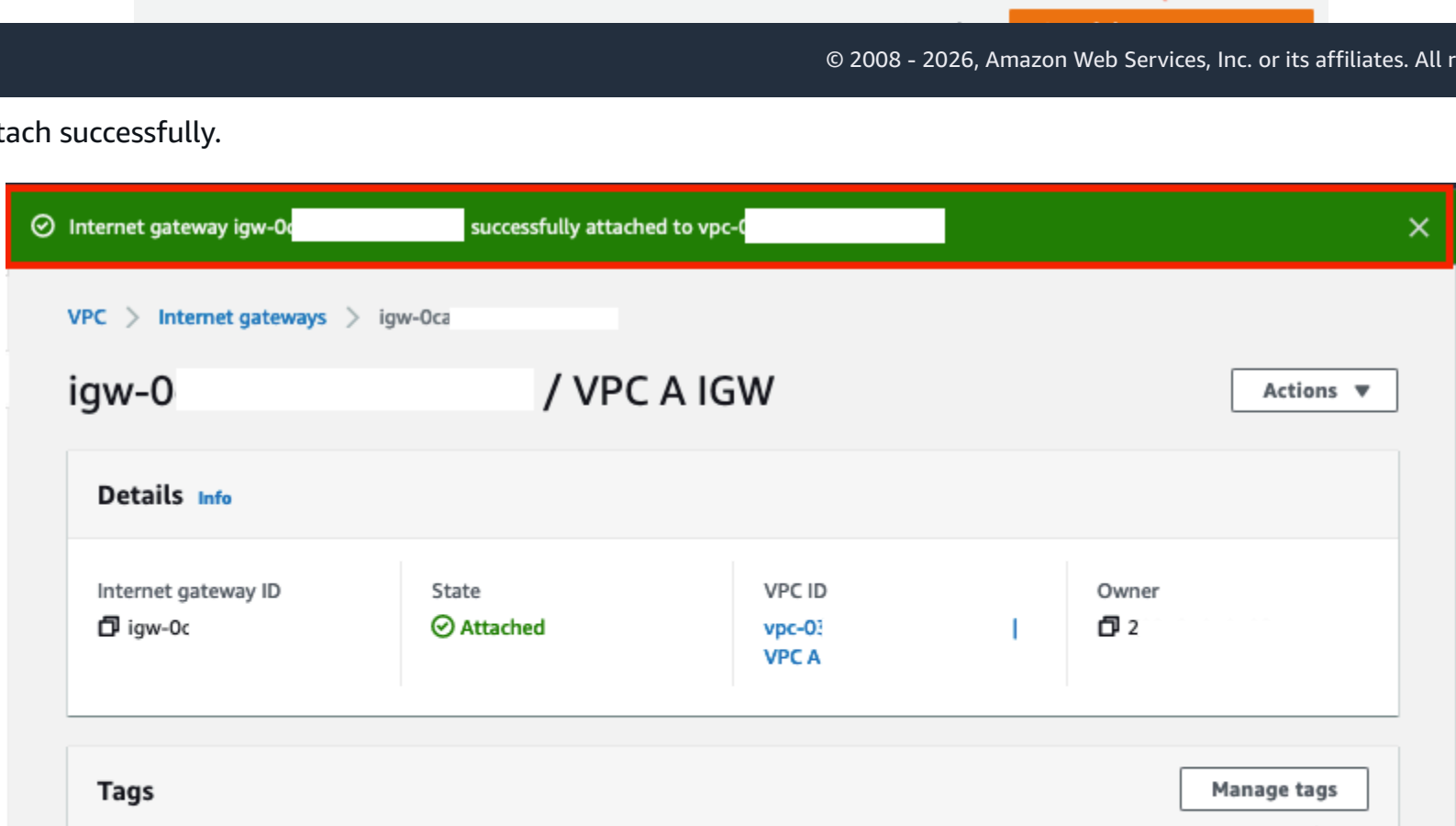
- On the success screen for the newly created IGW and click on [Attach to VPC](#):



- Select vpc-a from the dropdown list for Available VPCs and click [Attach internet gateway](#)



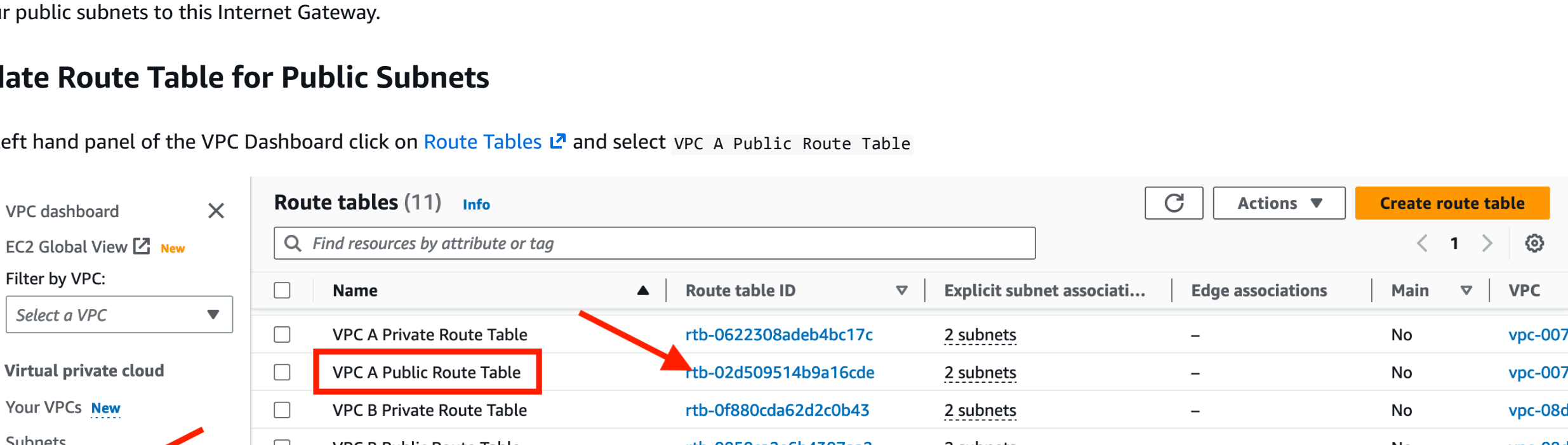
- The Internet Gateway should attach successfully.



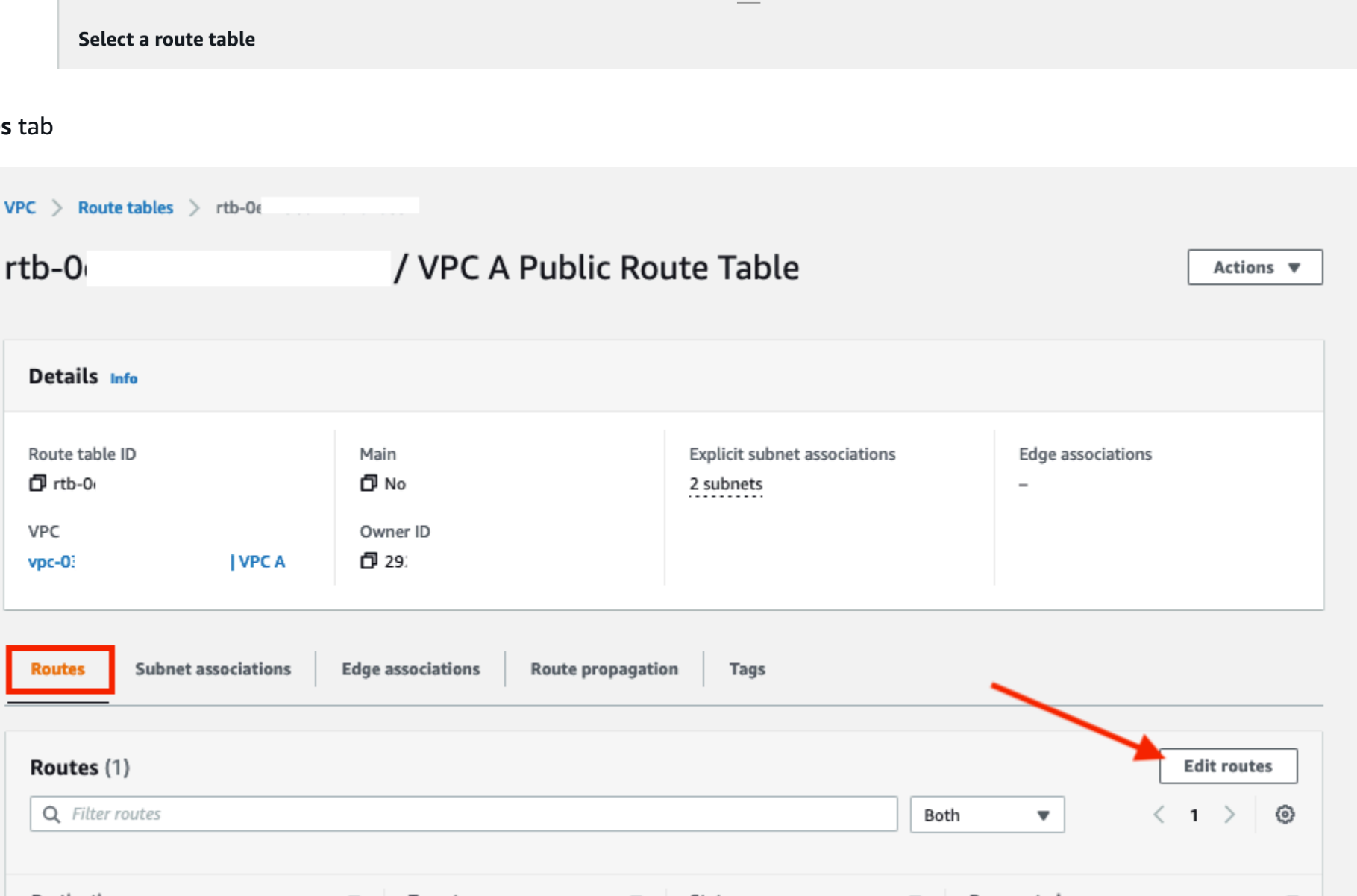
We now have an Internet access point for our VPC, but in order to utilize the newly created Internet Gateway, we need to update VPC routing tables to point the default routes for our public subnets to this Internet Gateway.

Update Route Table for Public Subnets

- In left hand panel of the VPC Dashboard click on [Route Tables](#) and select vpc-a Public Route Table



- Scroll down to the Routes tab

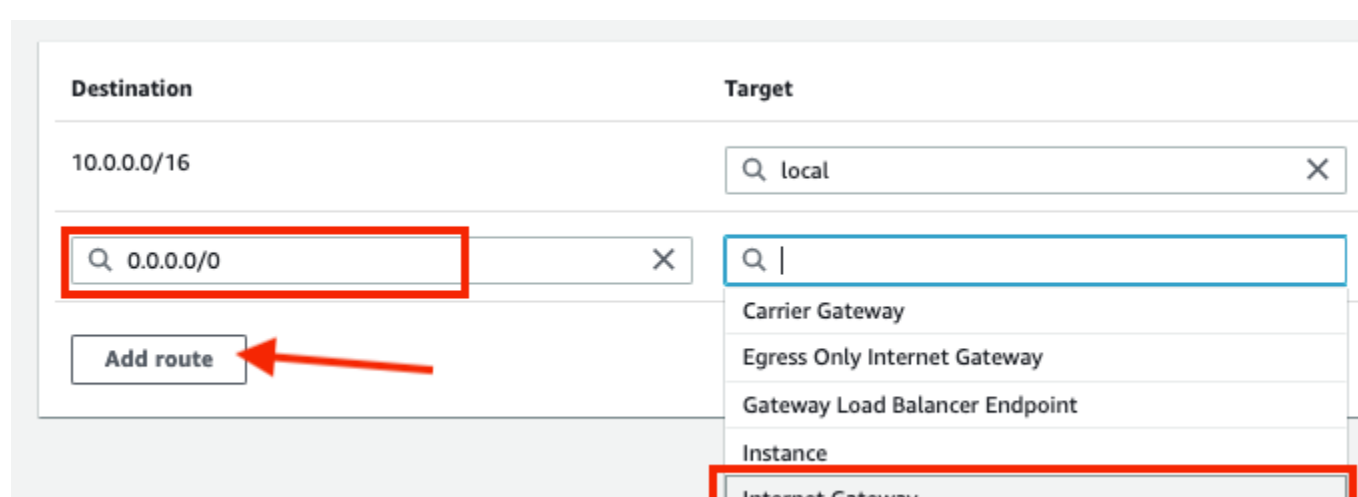


As you can see there is only a local route, so we're going to enable internet access by adding a route to the Internet Gateway

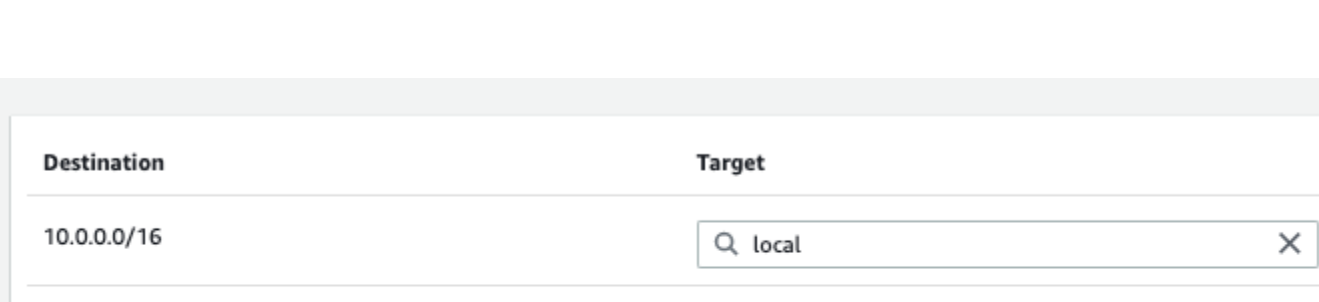
- Click on [Edit Routes](#)

- In the resulting screen

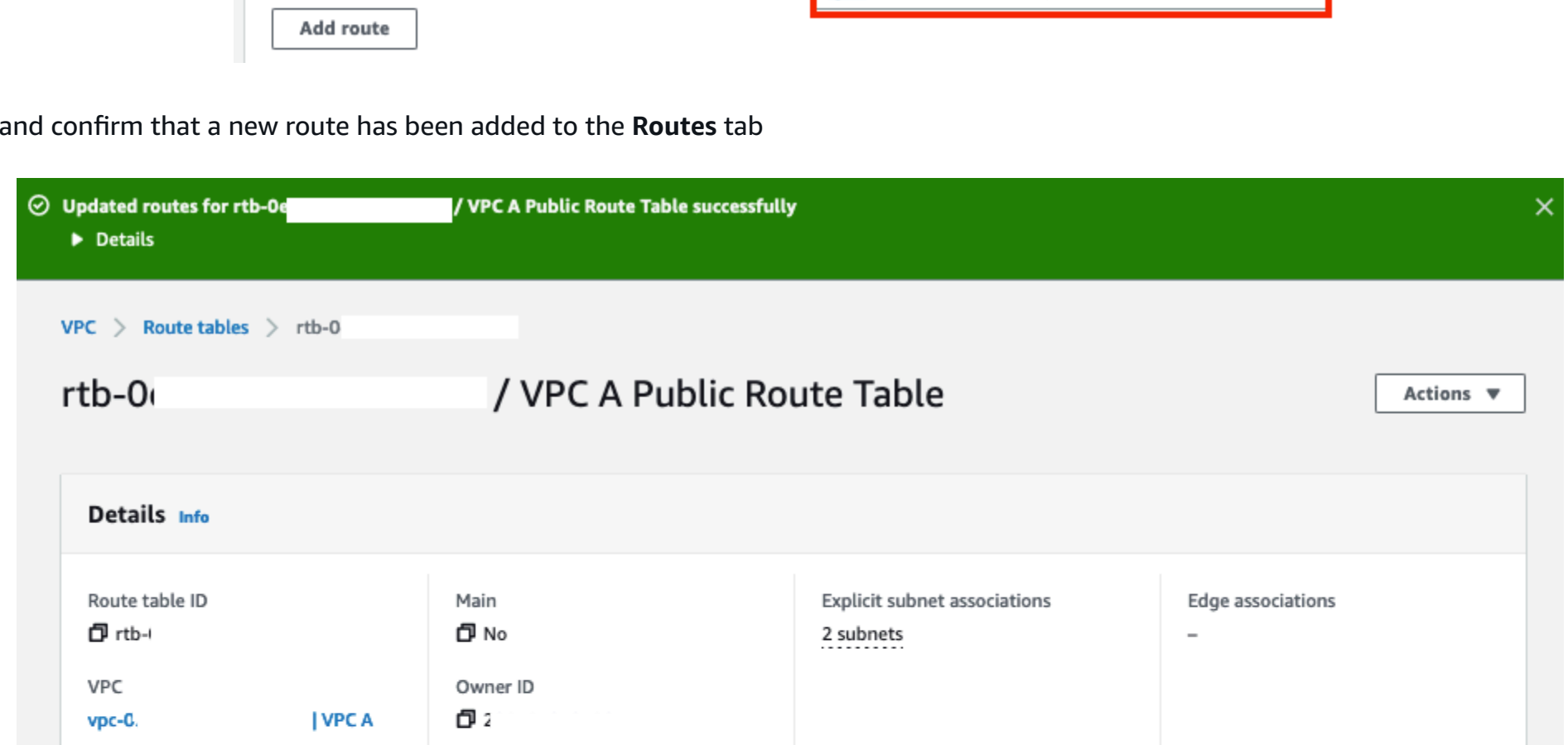
- Click on [Add route](#)
- Enter 0.0.0.0/0 in the [Destination](#)
- Select Internet Gateway from the [Target](#) dropdown



- Choose vpc-a-1gw



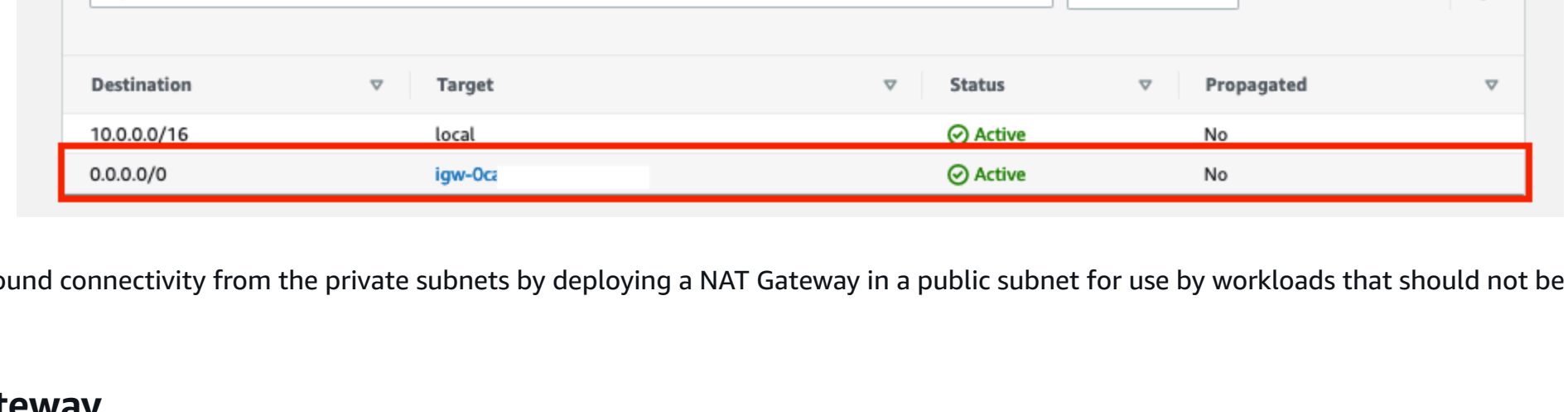
- Click [Save changes](#) and confirm that a new route has been added to the Routes tab



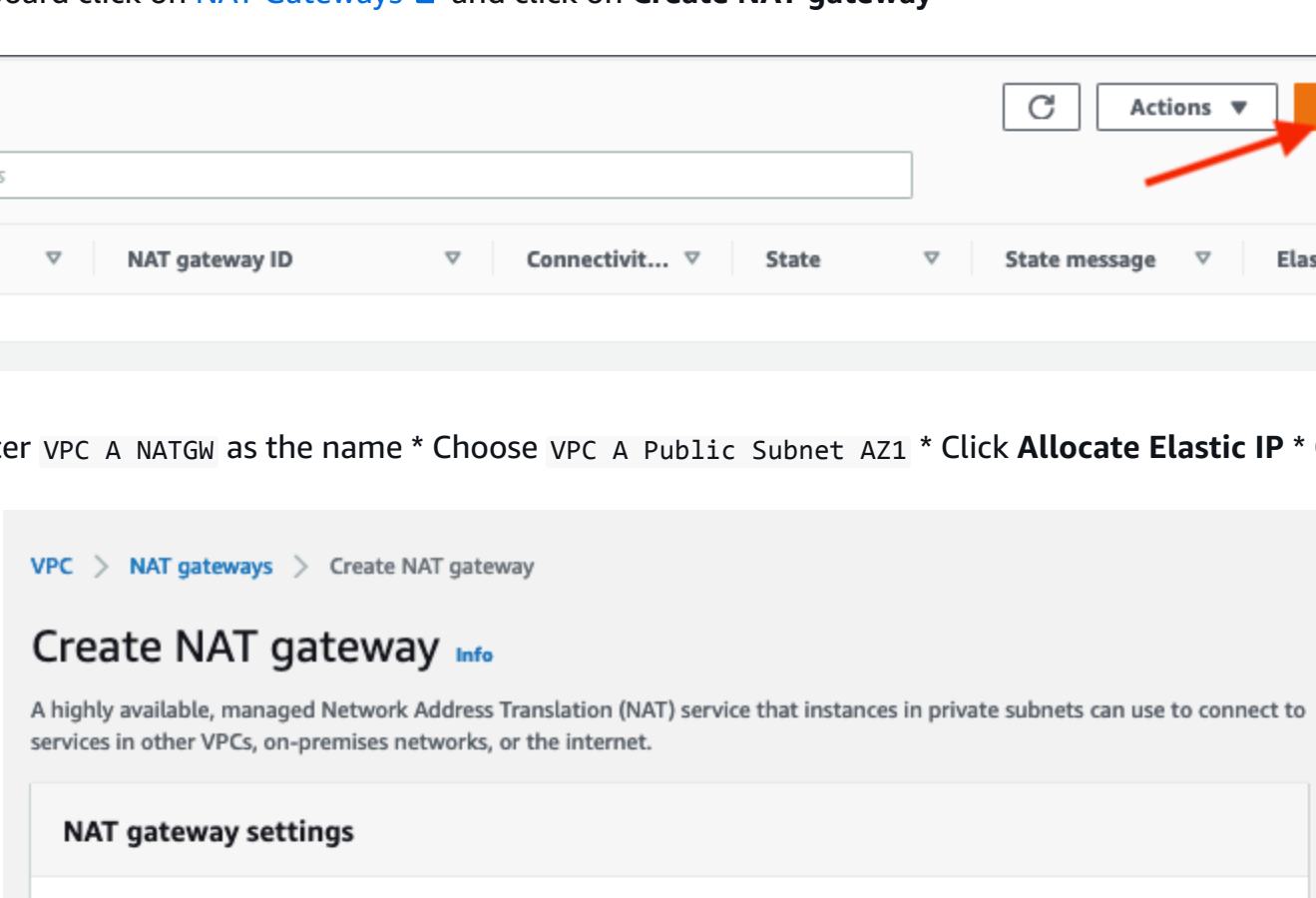
Next we will add outbound connectivity from the private subnets by deploying a NAT Gateway in a public subnet for use by workloads that should not be directly exposed to the internet.

Create NAT Gateway

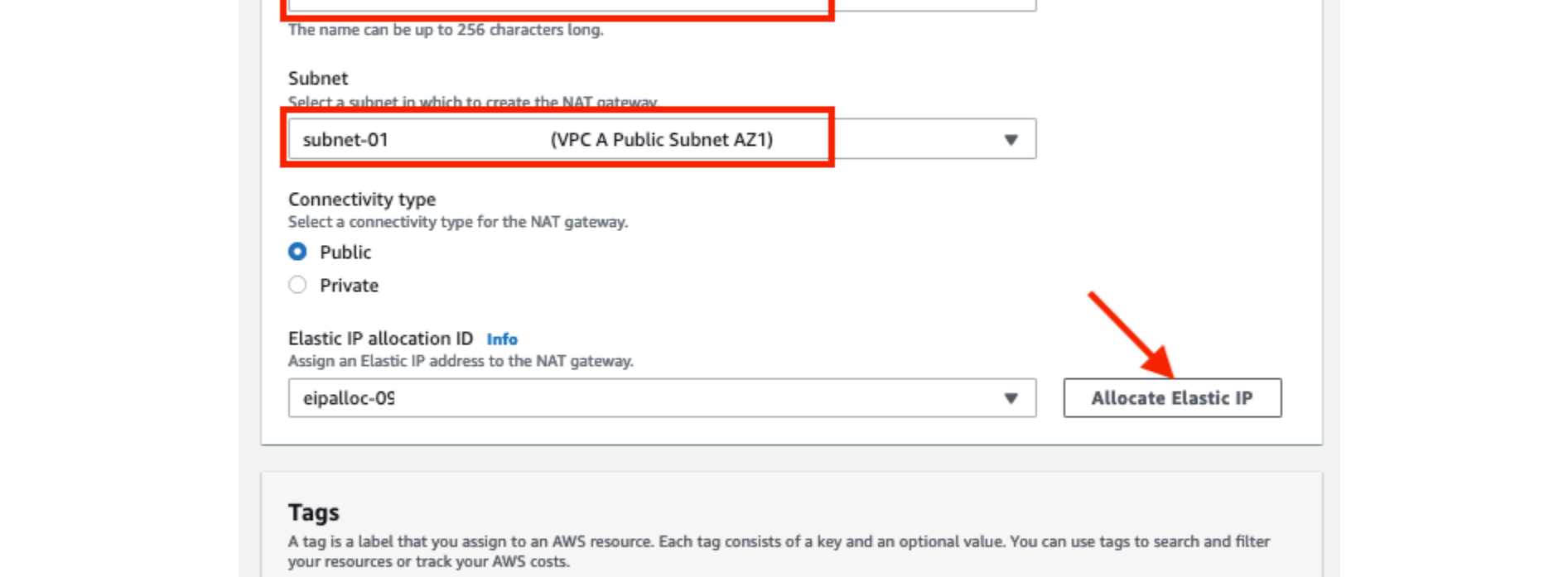
- In the left hand panel of the VPC Dashboard click on [NAT Gateways](#) and click on [Create NAT gateway](#)



- In the [Create NAT gateway](#) screen * Enter vpc-a-natgw as the name * Choose vpc-a Public Subnet AZ1 * Click [Allocate Elastic IP](#) * Click [Create NAT gateway](#)



- Upon creation the NAT Gateway details are displayed



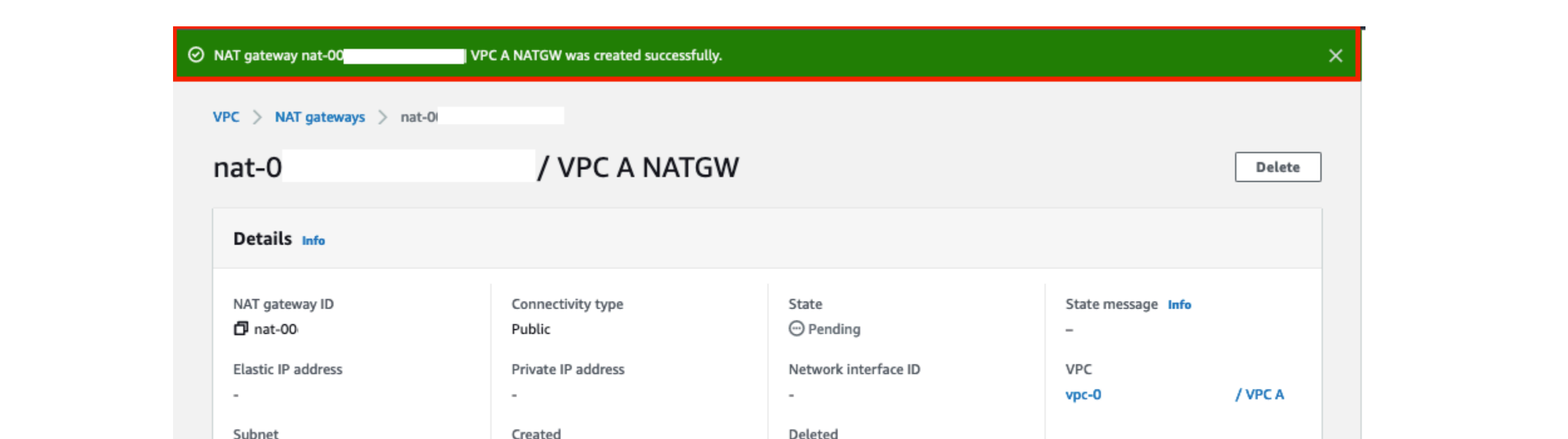
In this workshop, we only created one NAT Gateway in AZ1. It is best practice to create a NAT Gateway in each AZ that is utilized.

Update Route Table for Private Subnets

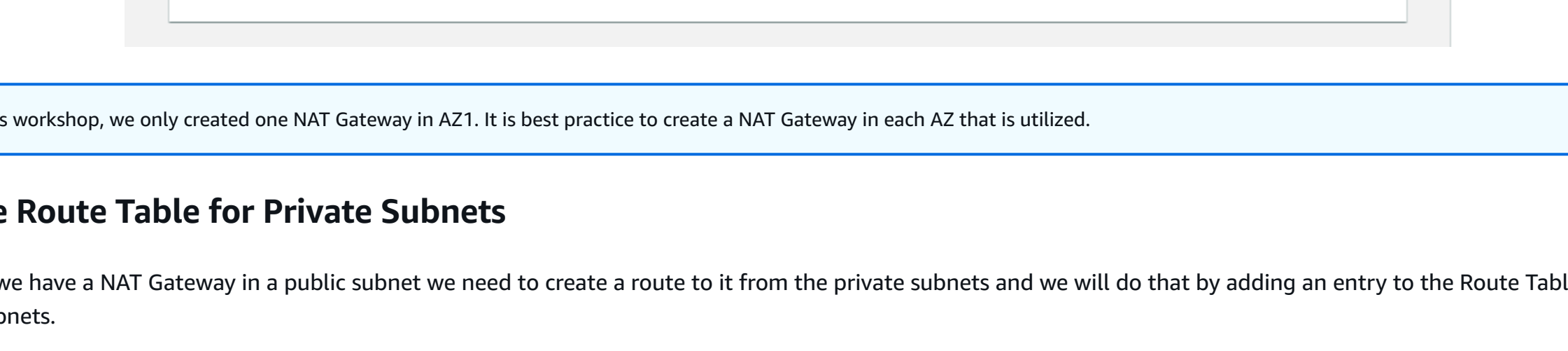
Now that we have a NAT Gateway in a public subnet we need to create a route to it from the private subnets and we will do that by adding an entry to the Route Table for the private subnets.

- In the left hand panel of the VPC Dashboard click on [Route Tables](#)

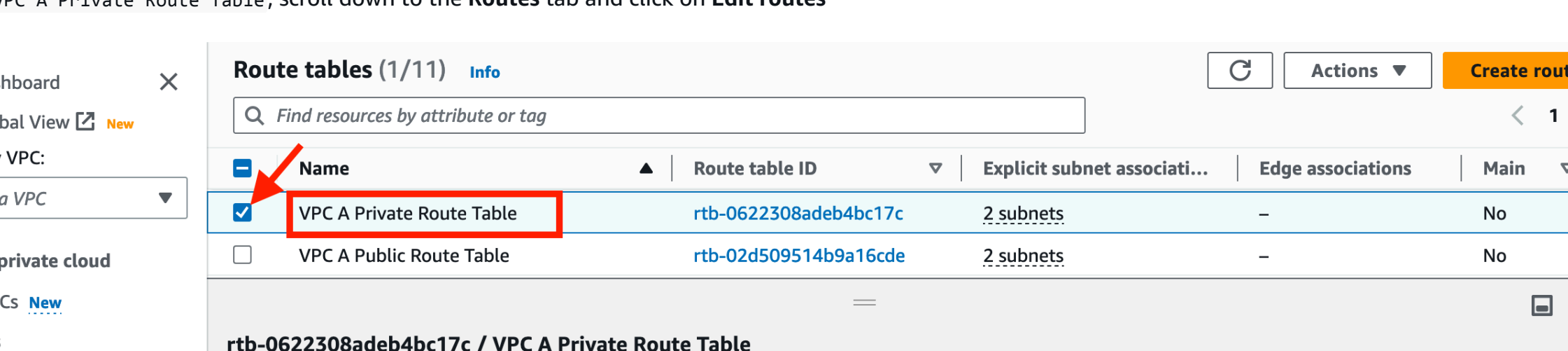
- Select vpc-a Private Route Table, scroll down to the Routes tab and click on [Edit routes](#)



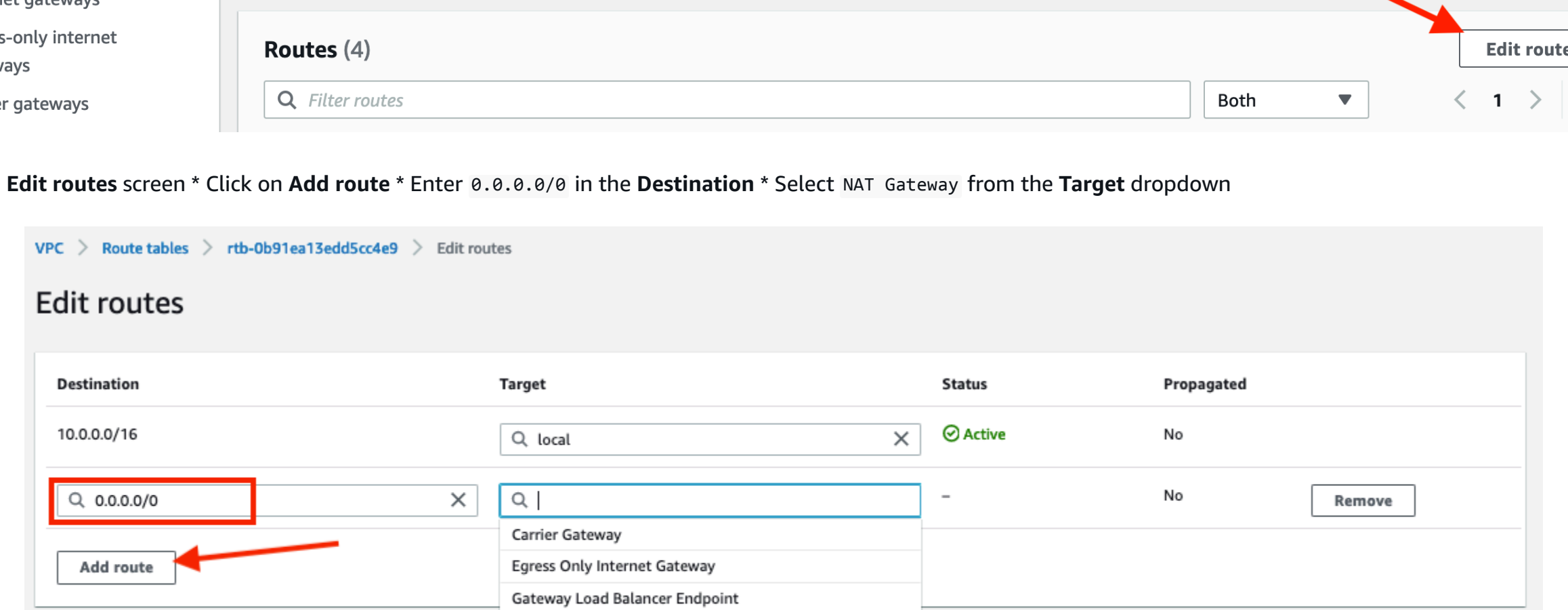
- In the [Edit routes](#) screen * Click on [Add route](#) * Enter 0.0.0.0/0 in the [Destination](#) * Select NAT Gateway from the [Target](#) dropdown



- Choose vpc-a-natgw and click on [Save changes](#)



- Confirm the new route appears in the Routes tab of the resulting screen



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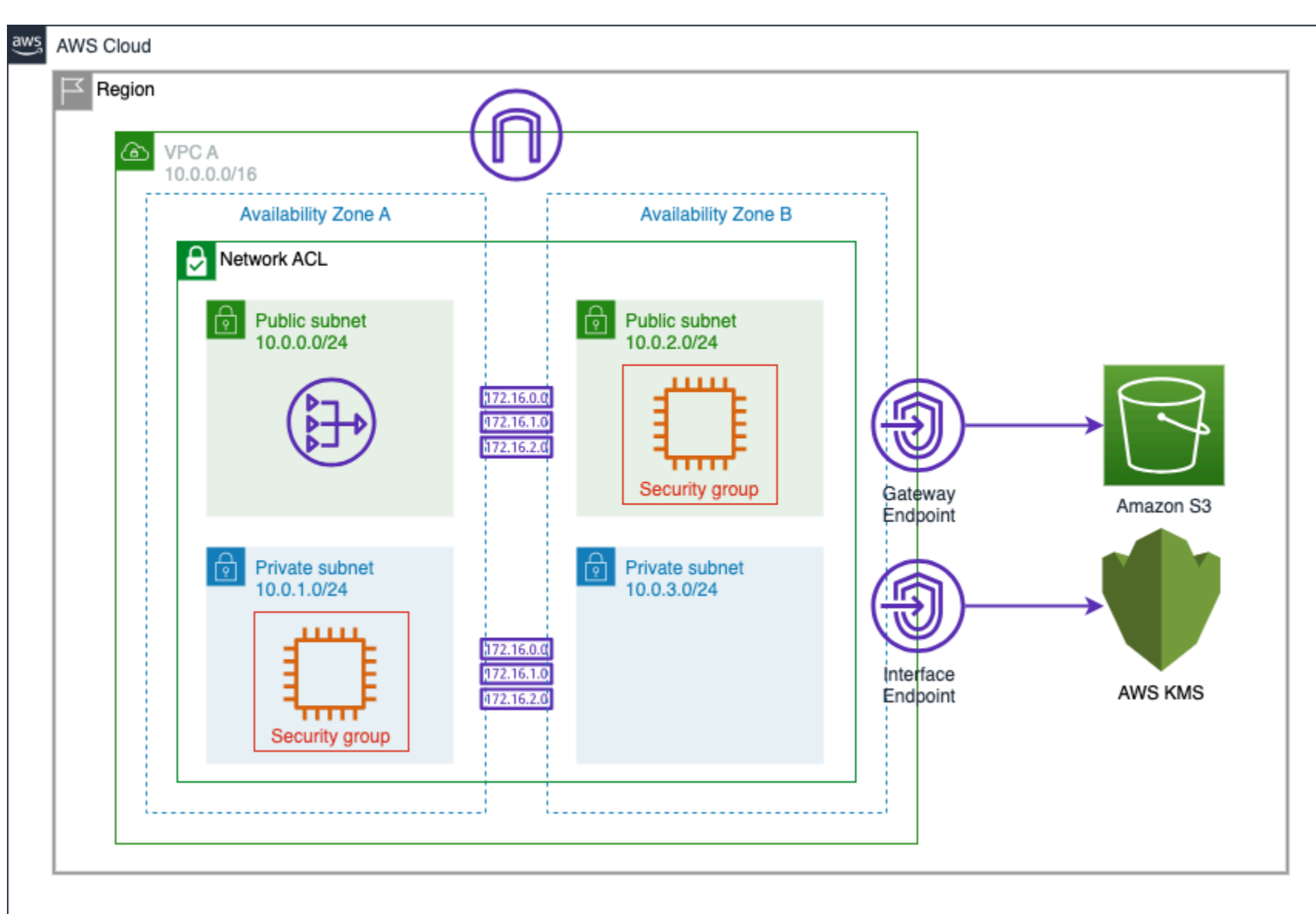
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Decline

Customize



1. Navigate to [Endpoints](#) with the VPC console and click on **Create Endpoint** to start creating a VPC Endpoint

The screenshot displays the AWS IAM console interface. On the left, a sidebar lists navigation options: Virtual private cloud, Your VPCs, Subnets, Route tables, Internet gateways, Egress-only Internet gateways, Carrier gateways, DHCP Option Sets, Elastic IPs, Managed prefix lists, **Endpoints** (highlighted with a red arrow), Endpoint services, NAT gateways, and Peering connections. The main content area is titled 'Endpoints' and includes a search bar with the placeholder 'filter endpoints'. Below the search bar is a table with the following columns: Name, VPC endpoint ID, VPC ID, Service name, Endpoint type, and Status. The table currently displays 'No endpoint found'. In the top right corner of the main content area, there are buttons for 'Actions' and 'Create endpoint' (highlighted with a red arrow).

2. In the **Endpoint settings** screen
 - Enter **VPC A KMS Endpoint** as the **Name tag**
 - Search for 'kms' under **Services**

Endpoint settings

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

VPC A KMS Endpoint

Service category

Select the service category

☒ AWS services
Services provided by Amazon

☐ AWS Marketplace services
Services that you've purchased through AWS Marketplace

☐ Private Link Ready partner services
Services with an AWS Service Ready designation

☐ Other endpoint services
Find services shared with you by service name

Services (181)

q kms

Service name	Owner	Type
Service name:amazonnews-us-east-1 kms	amazon	Interface
Service name:sagemaker-us-east-1-notebook		

3. From the results select the KMS service name without the '-fips' suffix

The screenshot shows the AWS IAM console 'Services' page. The filter 'Service Name: com.amazonaws.us-east-1-kms' is applied. The table lists the service details:

Service Name	Owner	Type
com.amazonaws.us-east-1-kms	amazon	Interface

4. In the **VPC** section
 - Select **VPC A** from the dropdown
 - Expand the **Additional settings** section
 - Ensure that **Enable DNS name** is checked
 - Select the **IPv4** radio button

VPC

Select the VPC in which to create the endpoint

The VPC in which to create your endpoint.

vpc-0a2f83322a1287eff (VPC A)

▼

⌕

▼

Additional settings

DNS name

☒

Enable DNS name

Info

Associates a private hosted zone with the VPC that contains a record set that enables you to leverage Amazon's private network connectivity to the service while making requests to the service's default public endpoint DNS name. To use this feature, ensure that the attributes Enable DNS hostnames and "Enable DNS support" are enabled for your VPC.

DNS record IP type

☒

IPv4

☐

IPv6

⌕

aws:ipaddress

5. Select **VPC A Private Subnet AZ1** and **VPC A Private Subnet AZ2** from the subnets and check the **IPv4** radio button.

Subnets (2/6) Info

Availability Zone	Subnet ID
us-east-1a (use1-a24)	subnet-0c39e528c4ab8192
us-east-1b (use1-az6)	subnet-053ac532c2948bb13
us-east-1c (use1-a21)	No subnet available
us-east-1d (use1-a22)	No subnet available
us-east-1e (use1-a23)	No subnet available
us-east-1f (use1-a25)	No subnet available

subnet-0c39e528c4ab8192 X VPC A Private Subnet A21

subnet-053ac532c2948bb13 X VPC A Private Subnet A22

IP address type

IPv4

IPv6

Dualstack

6. Select the default security group and leave the Policy as Full Access

Security groups (1/1) [Info](#)

< 1 >

☒	Group ID	Group name	VPC ID
☒	sg-024359c45a84fe4d8	default	vpc-0a2b8322a1287eff
sg-024359c45a84fe4d8			

Policy [Info](#)

VPC endpoint policy controls access to the service.

☒ **Full access**

Allow access by any user or service within the VPC using credentials from any Amazon Web Services accounts to any resources in this Amazon Web Services account. All policies — IAM user policies, VPC endpoint policies, and Amazon Web Services service-specific policies (e.g., Amazon S3 bucket policies, any S3 ACL policies) — must grant the necessary permissions for access to succeed.

☐ **Custom**

Use the [policy creation tool](#) to generate a policy, then paste the generated policy below.

- Click on **Create endpoint** button to create the VPC Endpoint for KMS in VPC A.
- Click on **Close** to return to the Endpoints screen.

Create a Gateway Endpoint for S3

1. Click 'Create Endpoint' to start creating another VPC Endpoint

Endpoints (1/1) info

Q Filter endpoints

VPC endpoint ID: vpc-e2f84460d130660bf Clear filters

Actions Create endpoint

☒	Name	VPC endpoint ID	VPC ID	Service name	Endpoint type	Status
☒	VPCA KMS Endpoint	vpc-e2f84460d130660bf	vpc-0a2f8b3322a1287...	com.amazonaws.us-east-1...	Interface	Pending

2. In the **Create endpoint** screen, enter `vpc A S3 Endpoint` search for 'S3' by service name

Endpoint settings

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

VPC A S3 Endpoint

Service category
Select the service category

- ☒ AWS services
Services provided by Amazon
- ☐ PrivateLink Ready partner services
Services with an AWS Service Ready designation
- ☐ AWS Marketplace services
Services that you've purchased through AWS Marketplace
- ☐ Other endpoint services
Find services shared with you by service name

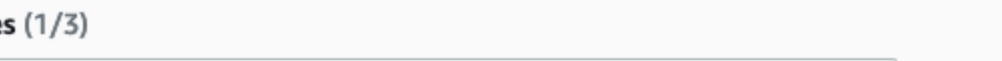
Services (181)

Search: s3

Service Name: com.amazonaws.s3-global-accesspoint

Service Name: com.amazonaws.us-east-1.s3

3. Select the endpoint that has a "Type" listed as "Gateway" and in the drop down box for VPC



The screenshot shows the AWS IAM console 'Services' page. A filter is applied to the 'Service Name' column, showing only services with 'com.amazonaws.us-east-1-s3' in their name. The table has two columns: 'Service Name' and 'Type'. The 'Type' column for all three services is 'Gateway'.

Service Name	Owner	Type
com.amazonaws.us-east-1-s3	amazon	Gateway
com.amazonaws.us-east-1-s3	amazon	Interface
com.amazonaws.us-east-1-s3.outposts	amazon	Interface

4. Select **VPC A** as the **VPC** and check the checkbox for all the route tables

VPC
Select the VPC in which to create the endpoint

VPC
The VPC in which to create your endpoint

vpc-0a2f83322a1287eff (VPC A)

Route tables (3/3)

Filter route tables

☒	Name	Route Table ID	Main
☒	VPC A Public Route Table	rtb-0e11186551a7ffc20 (VPC A Public ...)	No
☒	VPC A Private Route Table	rtb-05670455971df43ed (VPC A Privat...	No
☐	vpc-d1.6E75ha7c2a1286af	rtb-d1.6E75ha7c2a1286af	Yes

- ### 5. Leave the Bellows at 13 :

Policy info

VPC endpoint policy controls access to the service.

Full access

Allow access by any user or service within the VPC using credentials from any Amazon Web Services accounts to any resources in the Amazon Web Services service. All policies – IAM user policies, VPC endpoint policies, and Amazon Web Services service-specific policies (e.g. Amazon S3 bucket policy, any S3 ACL policies) – must grant the necessary permissions for access to succeed.

Custom

Custom policies must be explicitly created to restrict access to specific resources within the VPC endpoint policy.

6. Click on **Create endpoint** button to create the VPC Endpoint for S3 attached to VPC A

Successfully created VPC endpoint
vpc-0d30e06f2c54c34b

Endpoints (1/1) info

Filter endpoints

VPC endpoint ID: vpc-0d30e06f2c54c34b X Clear filters

☒	Name	VPC endpoint ID	VPC ID	Service name	Endpoint type	Status
☒	VPC & S3 Endpoint	vpc-0d30e06f2c54c34b	vpc-0a79f81e327a1787	com.amazonaws.us-east-1.s3	Gateway	Available

We have now gone through the bread and butter of AWS networking and built a networking foundation of public and private subnets across two availability zones with internet access and private connectivity to AWS service endpoints.

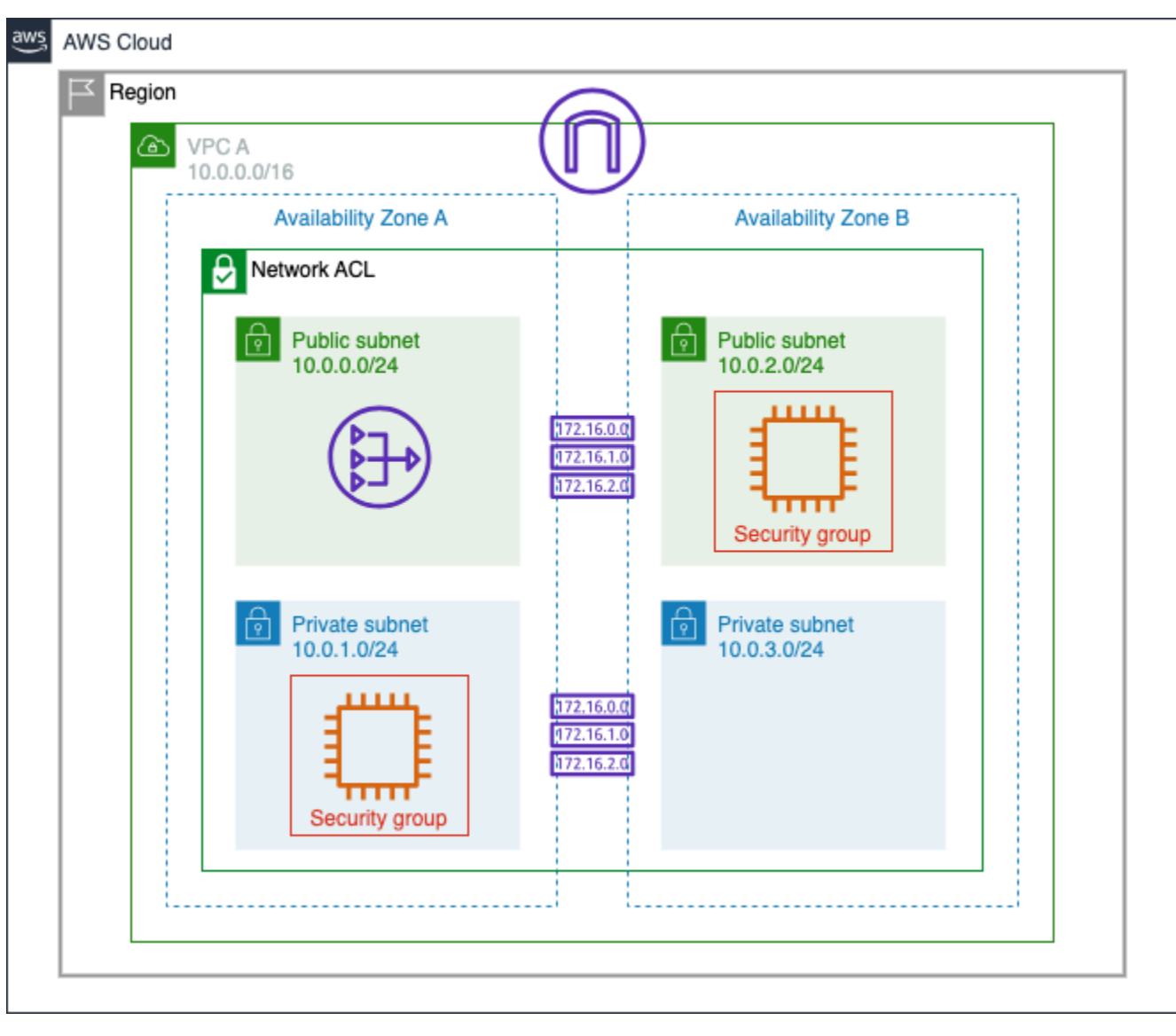
Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

- Accept
- Decline
- Customize

EC2 instances

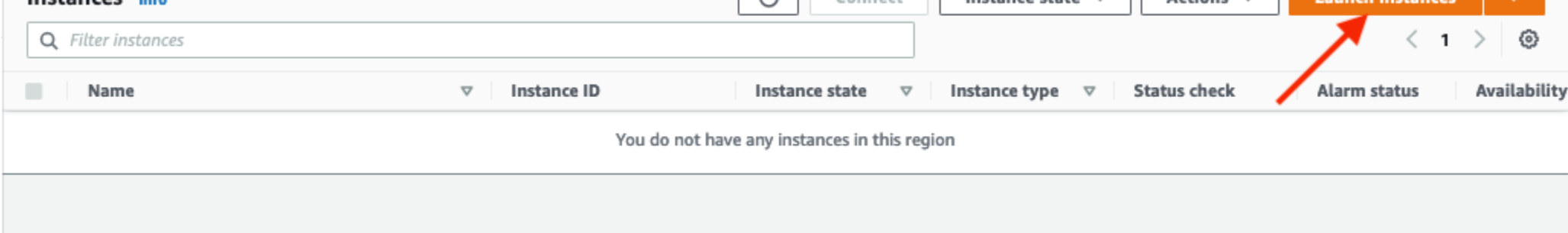
In this section, you will spin up EC2 instances in your VPC and protect them with a security group only allowing ICMP traffic to reach the hosts.



Launch an EC2 Instance into a Public Subnet

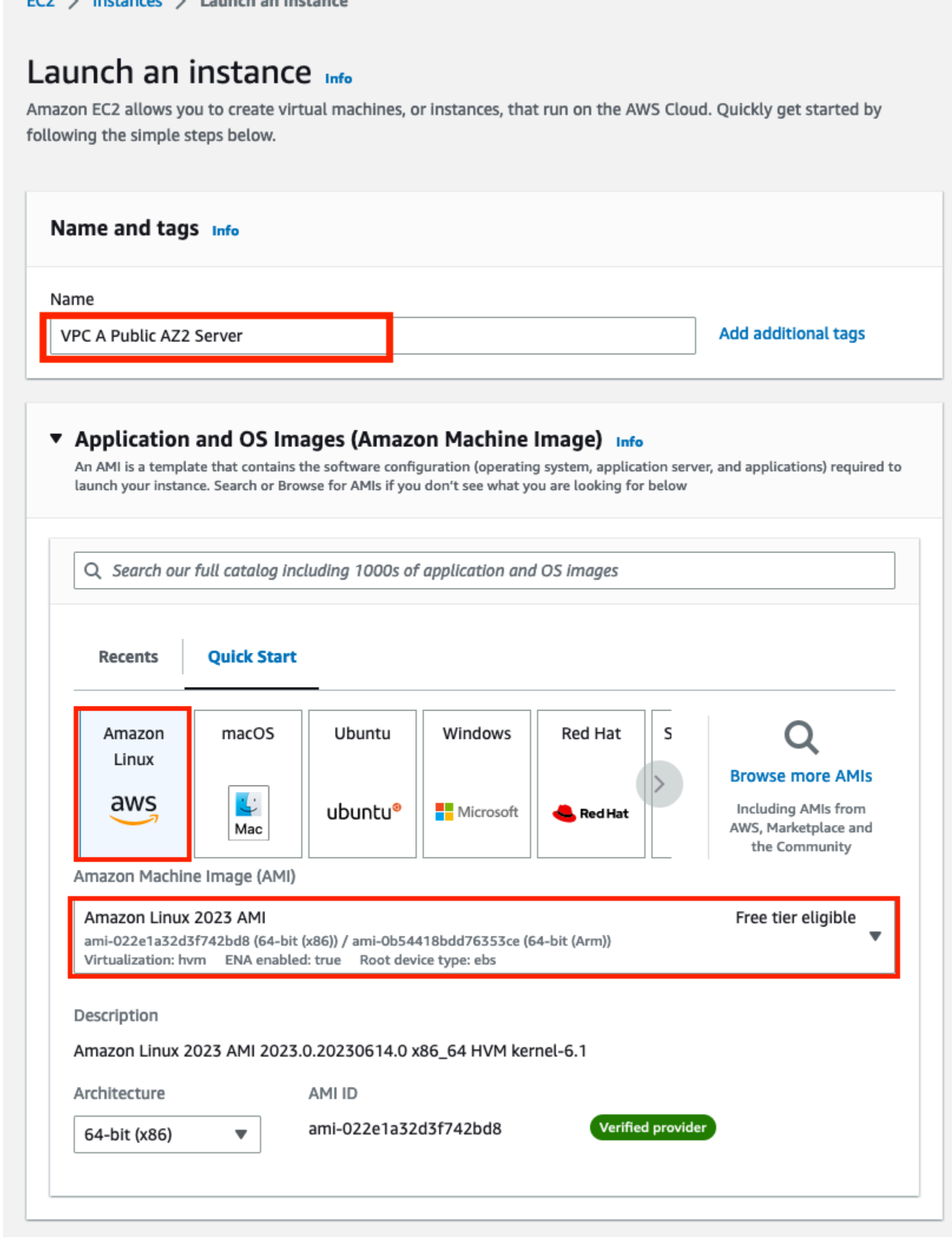
In this section, you will create an EC2 instance in the Public Subnet of AZ2 (Availability Zone B). You will create the instance in AZ1 in the next section.

- In the [Instances](#) section of the EC2 console click **Launch Instances**

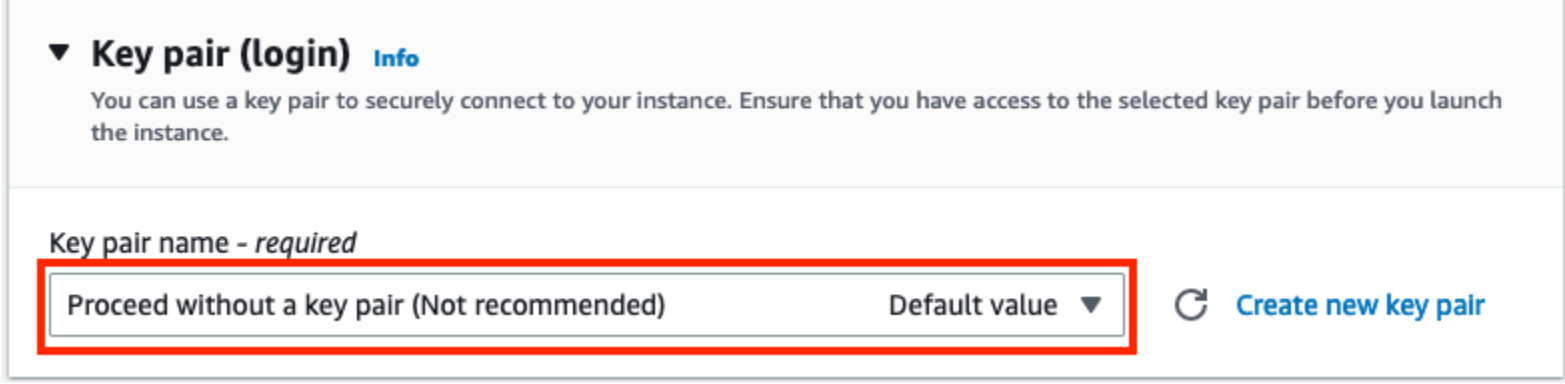


- In the resulting **Launch an instance** screen

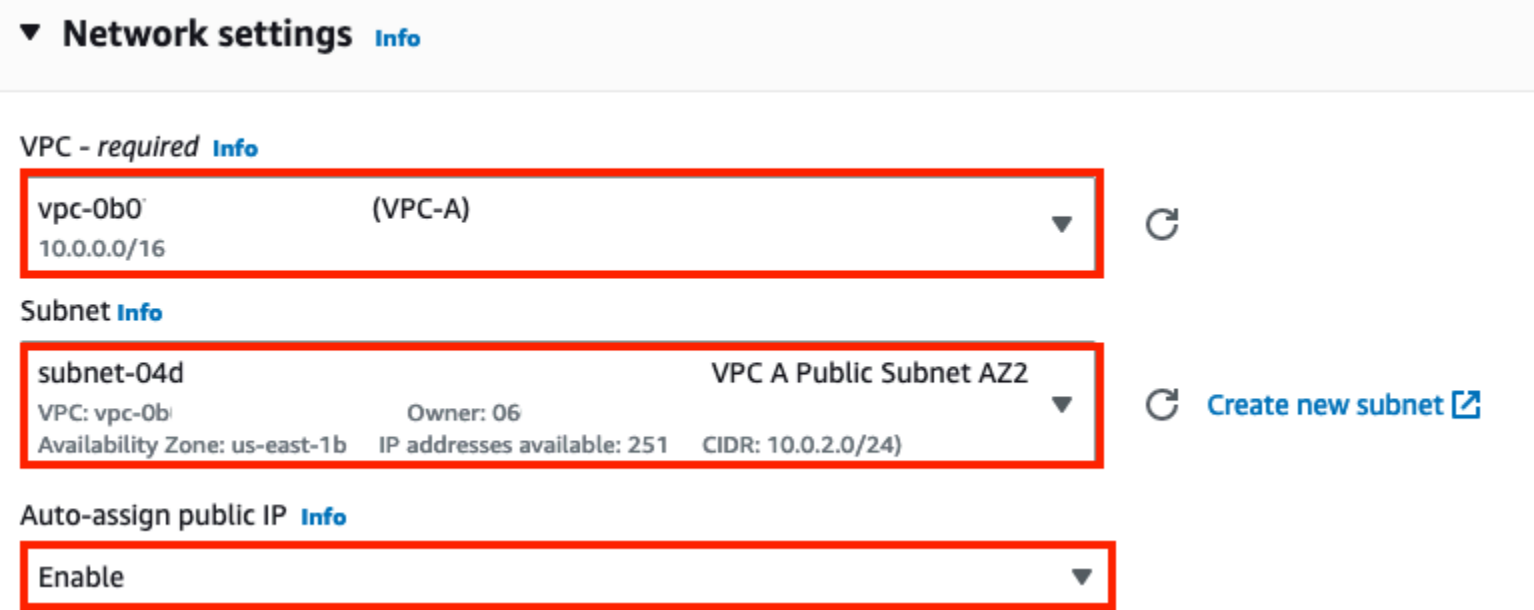
- Enter VPC A Public AZ2 Server for the **Name**
- Ensure that Amazon Linux 2023 AMI will be selected, and the instance type is t2.micro.



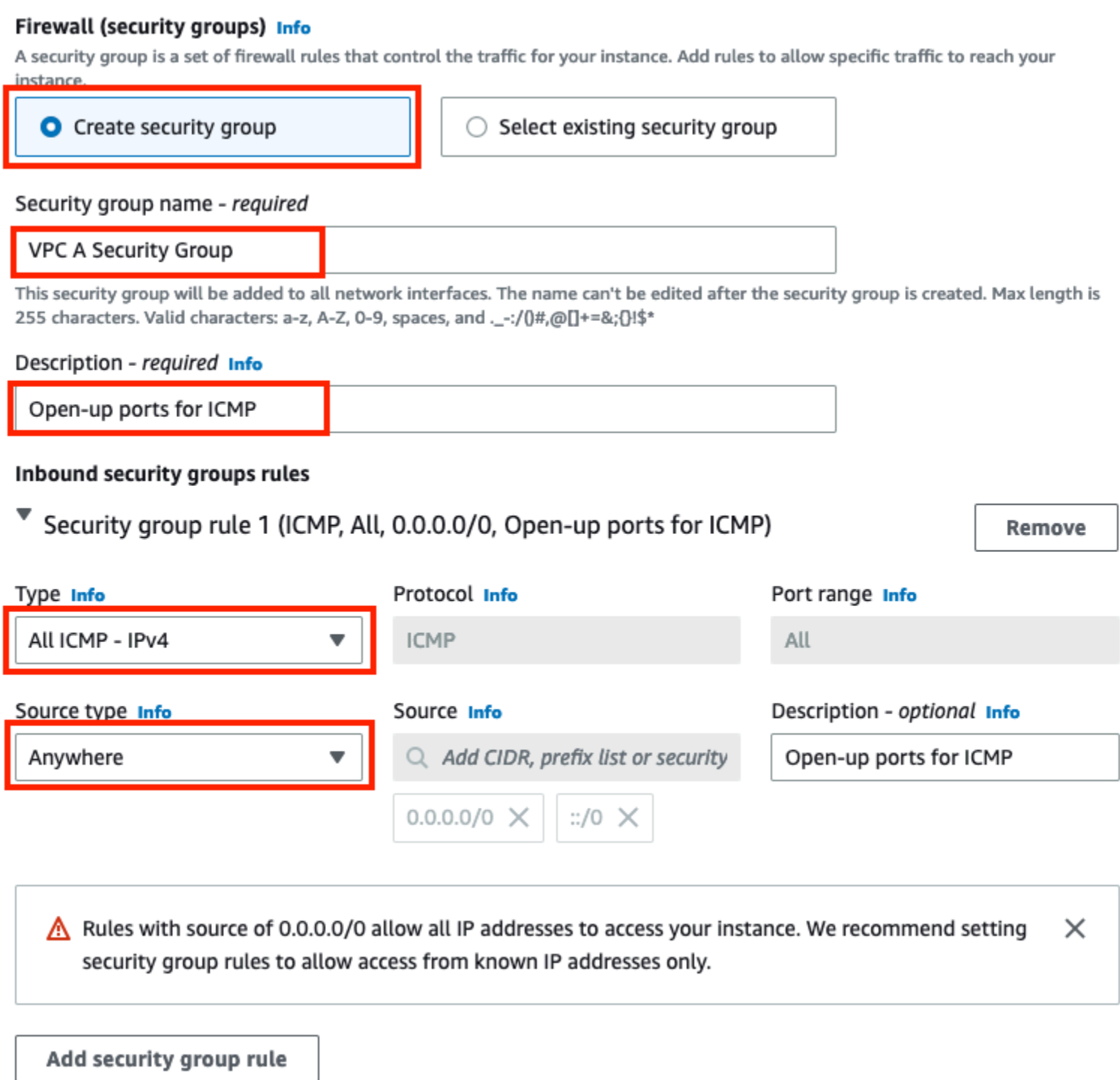
- Under **Key pair (login)** select **Proceed without a key pair**. A key pair is not needed since we will be using Systems Manager to connect to the instances.



- Under **Network settings** click **Edit** and
 - Select VPC A from the dropdown for the VPC field
 - Select VPC A Public subnet AZ2 from the dropdown for the Subnet field

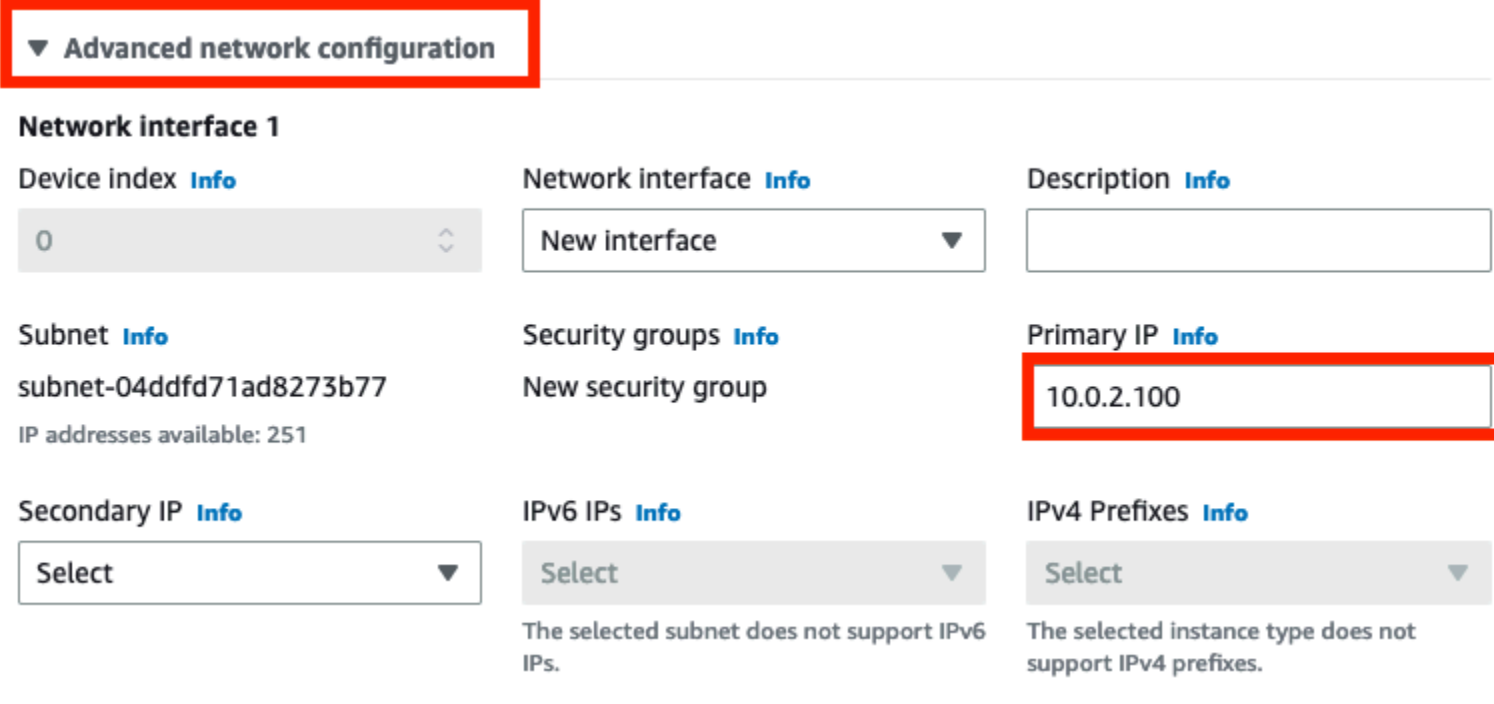


- Select **Create security group** with the name VPC A Security Group, description of open-up ports for ICMP
- In **Inbound security groups rules** under **Type** select All ICMP - IPv4 and enter 0.0.0.0/0 as the Source

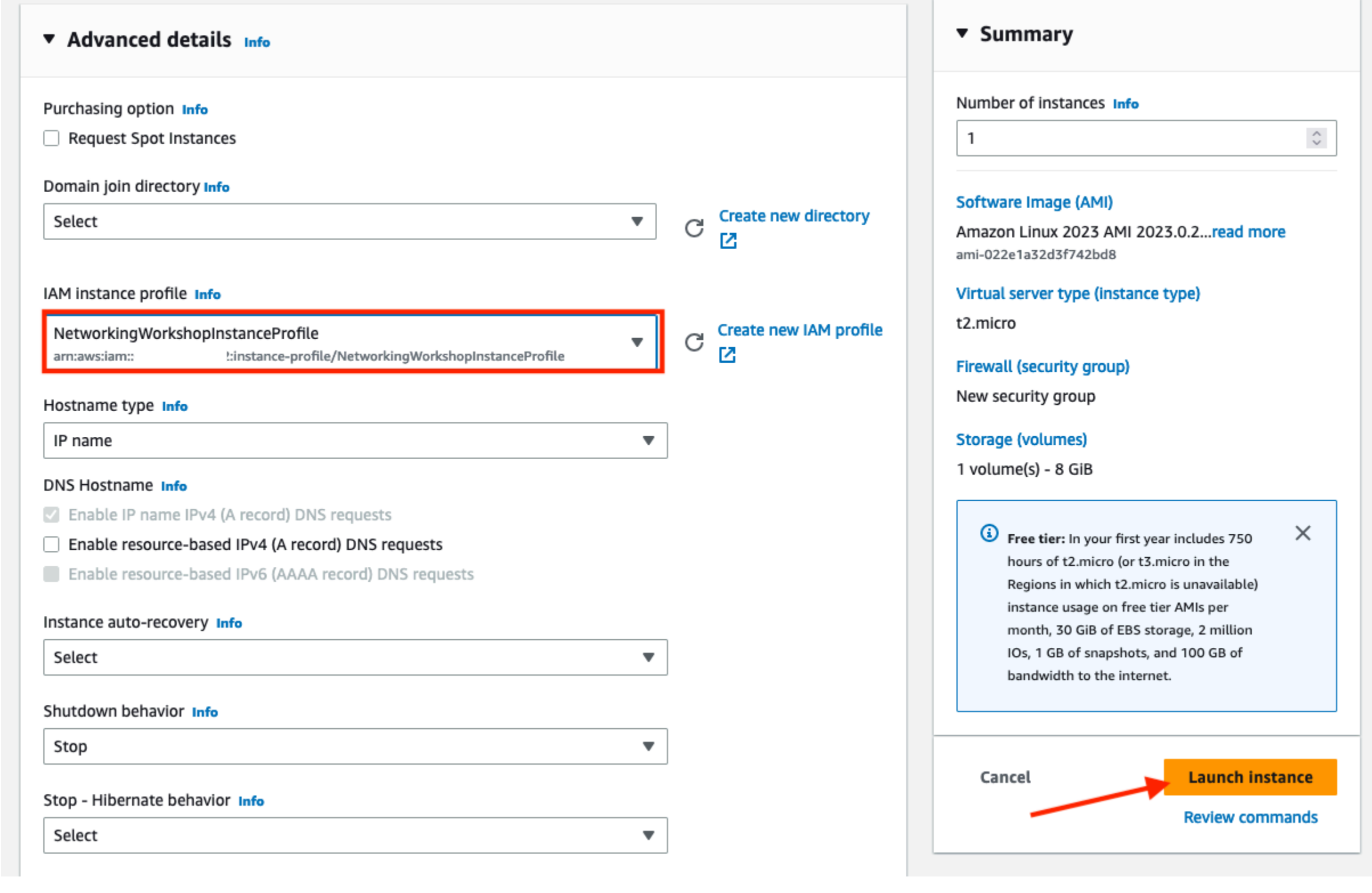


Since security groups are stateful, you don't need to edit the outbound rules. The security group will allow the instance to respond to the ping since it saw the ping arrive at the instance.

- Expand **Advanced network configuration** and under **Primary IP** enter 10.0.2.100.



- At the bottom of the section
 - Expand **Advanced details**
 - Under **IAM instance profile** select **NetworkingWorkshopInstanceProfile** which was created in the pre-requisites section.
 - Click **Launch instance**



- Click **View all instances**

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
EC2 Dashboard						
EC2 Global View	VPC A Public AZ2 Server	i-085ff1a3fd67cce	Running	t2.micro	2/2 checks passed	No alarms
Events						

If you just finished the last step, your EC2 instance might still be spinning up. You can tell by looking at the Instance State and Status Checks columns. If you see **Pending** state or status **Initializing**, the instance is not ready yet. After few minutes, you should have an EC2 instance in the "running" state.

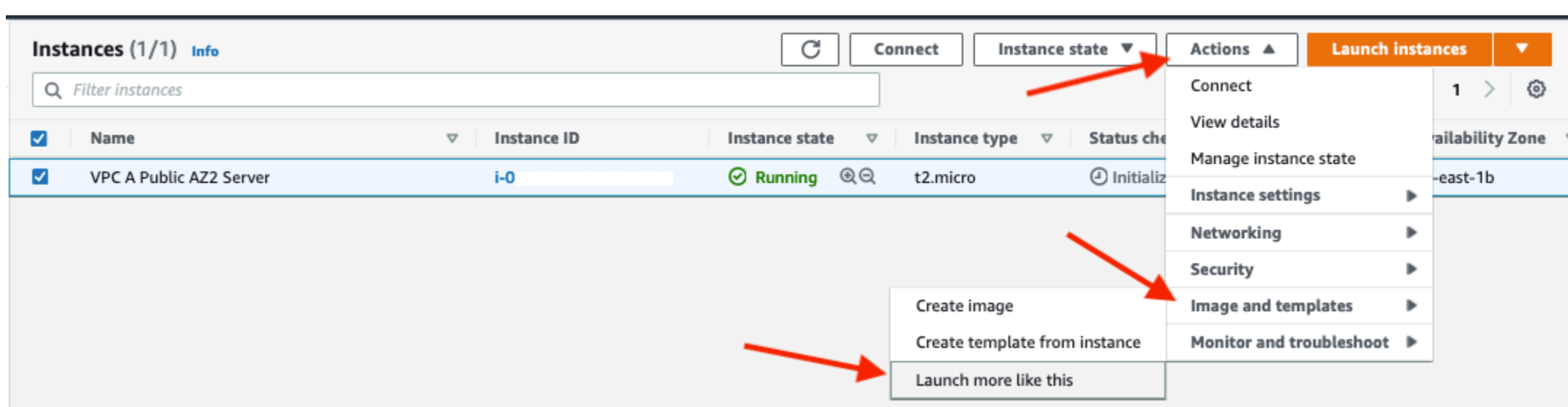
Congratulations! You have just launched a virtual server in your public subnet in AZ2.

Launch Instance in Private Subnet

You could follow the same process in the last two sections in order to deploy an EC2 instance into a private subnet, however it is also possible to launch a new instance using the same settings as previously.

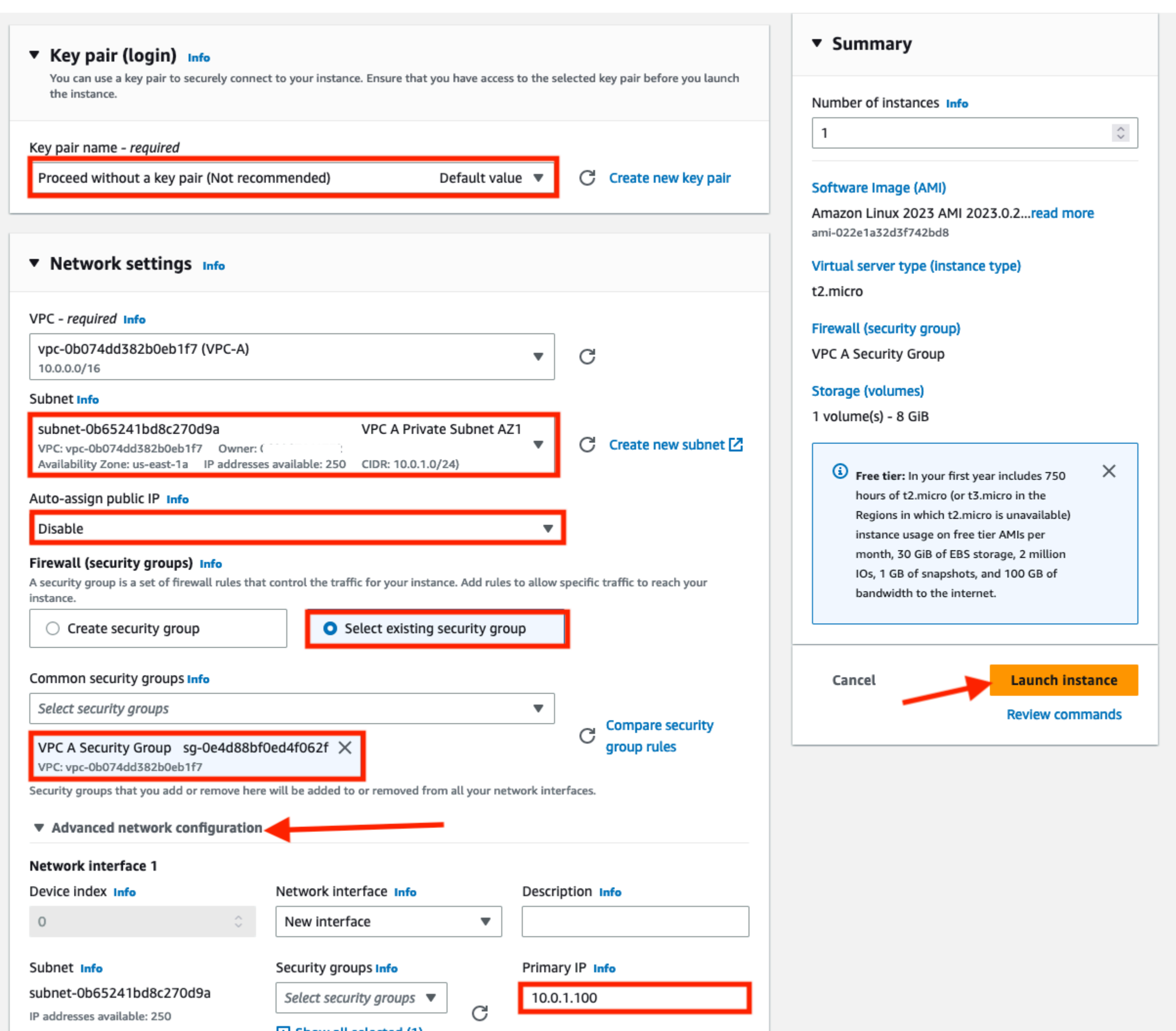
- In the [Instances](#) section of the EC2 console

- Select the running public instance VPC A Public AZ2 Server
- Click **Actions** then **Image and templates** then **Launch more like this**



- In the settings screen

- Update the **Name** to VPC A Private AZ1 Server
- Under **Key pair (login)** select **Proceed without a key pair**.
- Update the **Subnet** to be VPC A Private Subnet AZ1
- Set the **Auto-Assign Public IP** setting to **Disable**.
- Expand **Advanced network configuration** and under **Primary IP** enter 10.0.1.100.
- Click **Launch instance**



- Click **View all instances**. There should now be two EC2 instances running in the VPC

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
EC2 Dashboard	VPC A Public AZ2 Server	i-023dcb8af275e1c8	Running	t2.micro	2/2 checks passed	No alarms	us-east-1b
EC2 Global View	VPC A Private AZ1 Server	i-0c9b0b01c52ba8867	Running	t2.micro	-	No alarms	us-east-1a
Events							

AWS Networking Workshop

Foundational Topics

Prerequisites

VPC Fundamentals

Amazon VPC

Subnets

Network ACLs

Route Tables

Internet Connectivity

VPC Endpoints

EC2 Instances

Test Connectivity

Clean Up

Multiple VPCs

Security Controls

Connecting to On-Premises

Connecting to SDWAN

Network Monitoring

Network analytics using Amazon Quick Suite

Advanced Topics

Contributors & Revision History

Workshop catalog in AWS Builder Center

Content preferences

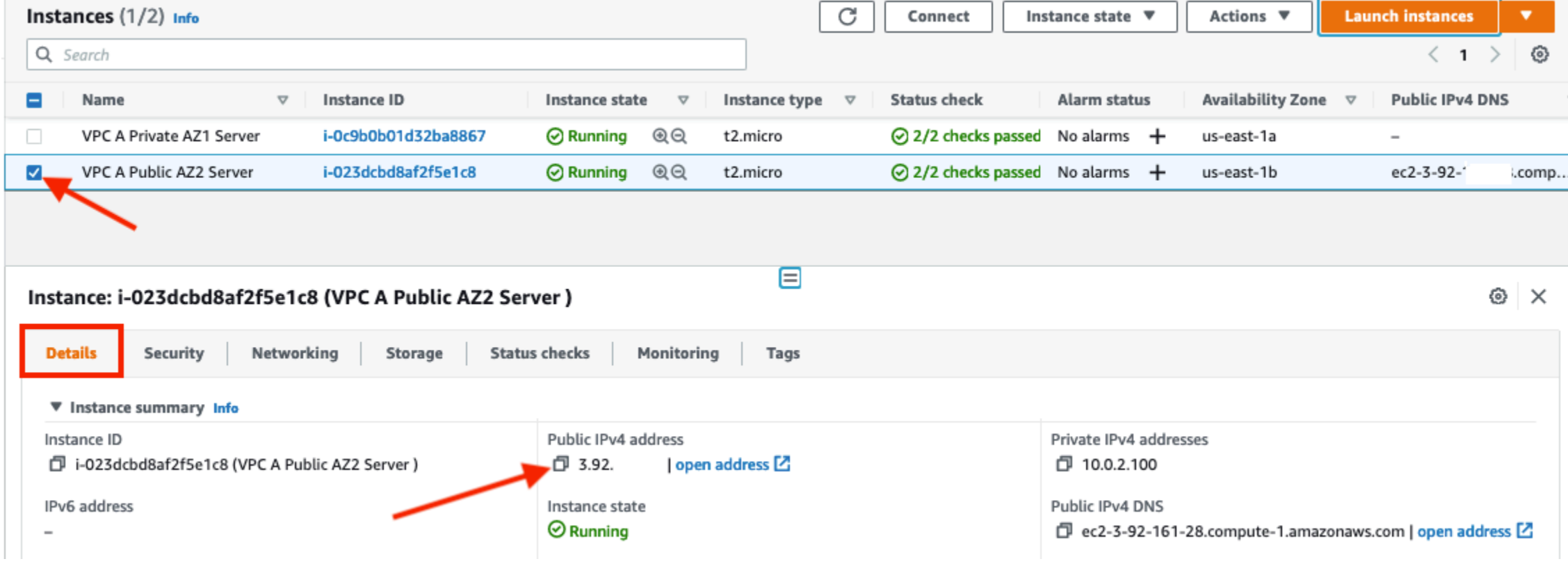
Language

English

Test Connectivity

In this section, you will use the EC2 instances in the public and private subnets of your VPC to test the connectivity for the network foundations you created in the previous section.

- In the [Instances](#) section of the EC2 console
 - Select the VPC A Public AZ2 Server instance
 - Scroll down to the **Details** tab
 - Copy the **Public IPv4 address** field by clicking the copy icon to the left of it

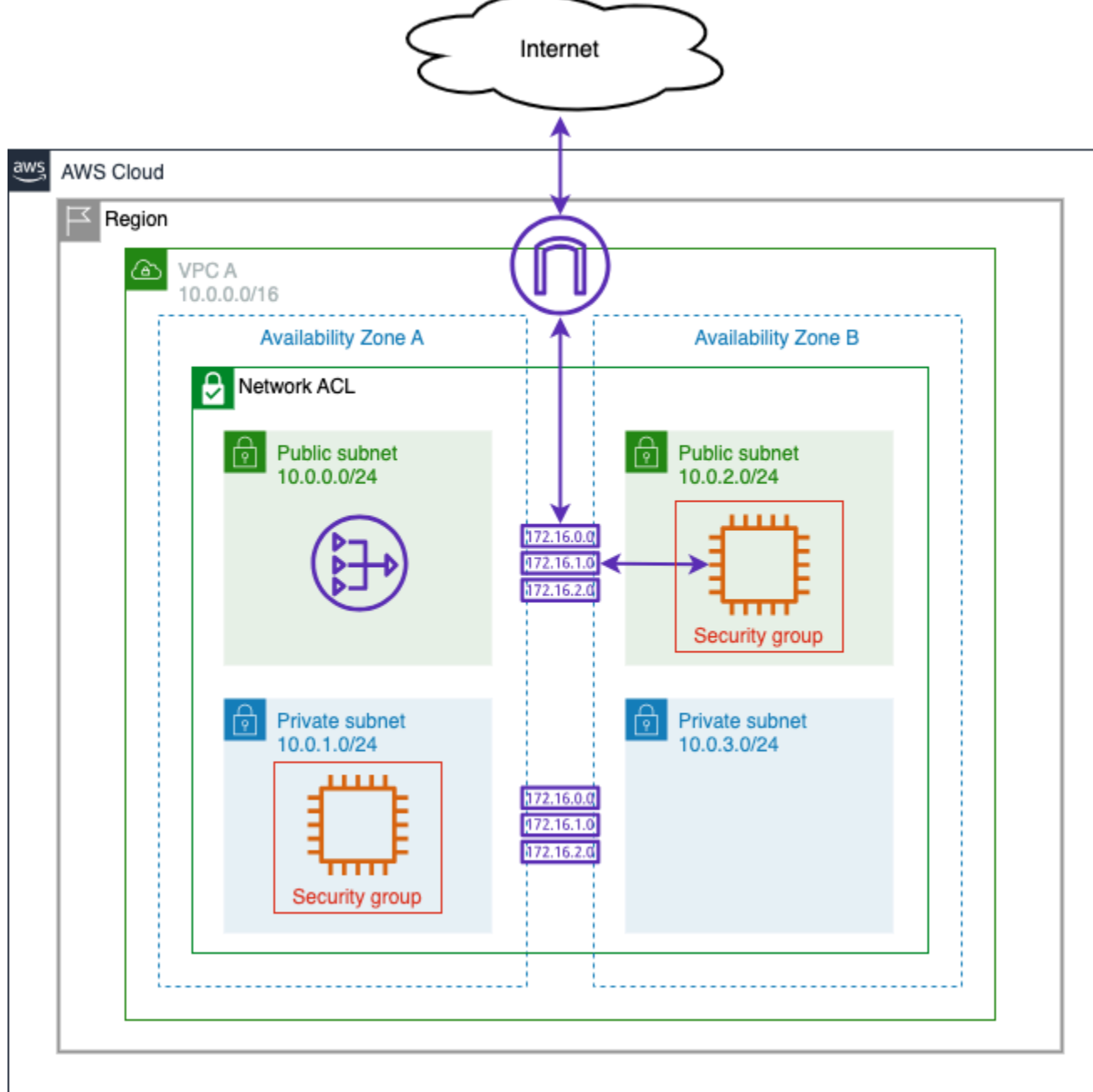


- To ping the instance, you need to open your CLI. On Windows, open the Command Prompt. On Mac, open the Terminal.
- Type `ping` then a space, then paste the Elastic IP from above, then space then `-c 5` and enter.

If the instance is reachable, we expect to see lines appearing such as

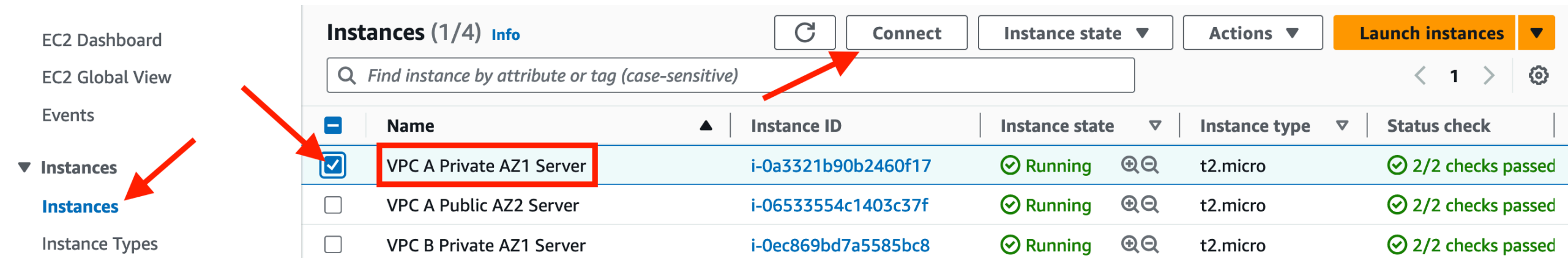
```
sh-4.2$ ping 54.211.142.109
PING 54.211.142.109 (54.211.142.109) 56(84) bytes of data.
 64 bytes from 54.211.142.109: icmp_seq=1 ttl=254 time=0.384 ms
 64 bytes from 54.211.142.109: icmp_seq=2 ttl=254 time=0.474 ms
 64 bytes from 54.211.142.109: icmp_seq=3 ttl=254 time=0.469 ms
 64 bytes from 54.211.142.109: icmp_seq=4 ttl=254 time=0.420 ms
 64 bytes from 54.211.142.109: icmp_seq=5 ttl=254 time=0.486 ms
--- 54.211.142.109 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4098ms
 rtt min/avg/max/mdev = 0.384/0.446/0.486/0.044 ms
sh-4.2$
```

Good job! You have successfully confirmed connectivity between the public EC2 instance and the internet.

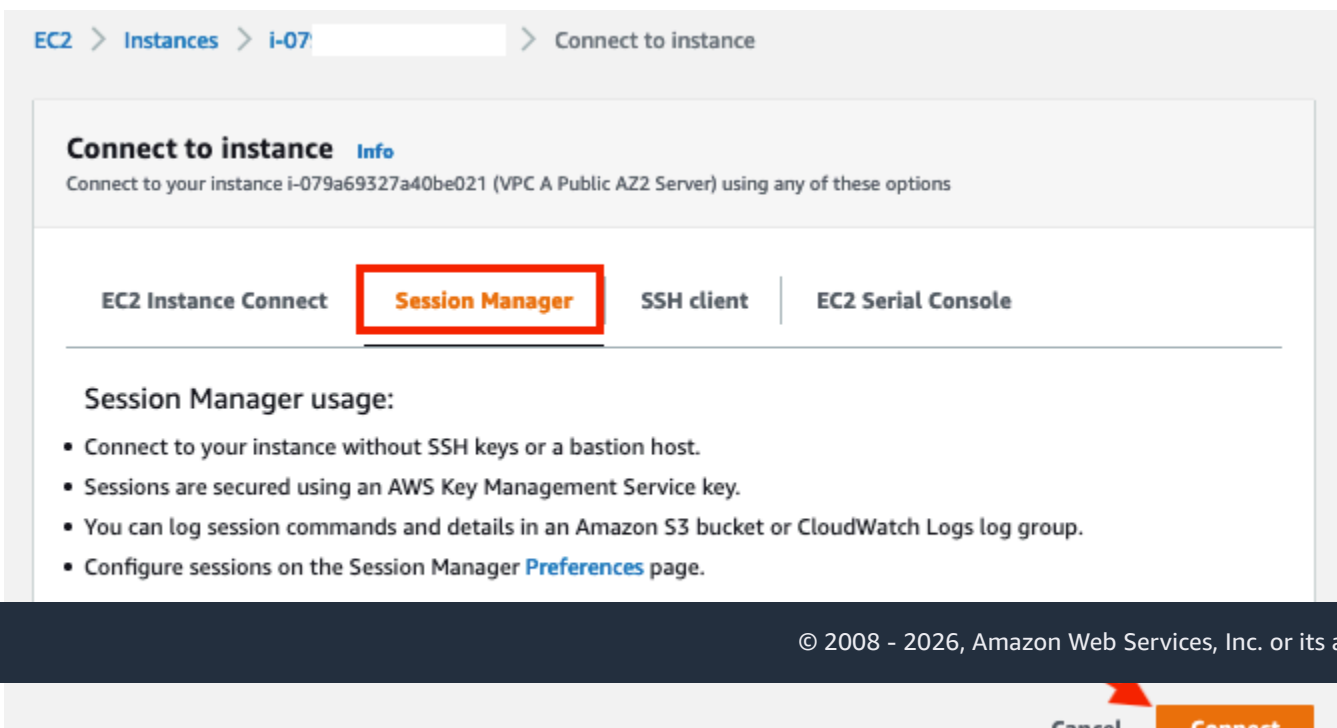


Test Private Instance Connectivity

- In the [Instances](#) section of the EC2 console
 - Select the VPC A Private AZ1 Server instance
 - Click on the **Connect** button



- In the following screen, select the Session Manager tab and click **Connect**



This will open a terminal from which you can test connectivity to both the public instance at 10.0.2.100 and external connectivity to `example.com` via the NAT Gateway.

- Copy the following ping commands and paste in the Session Manager console

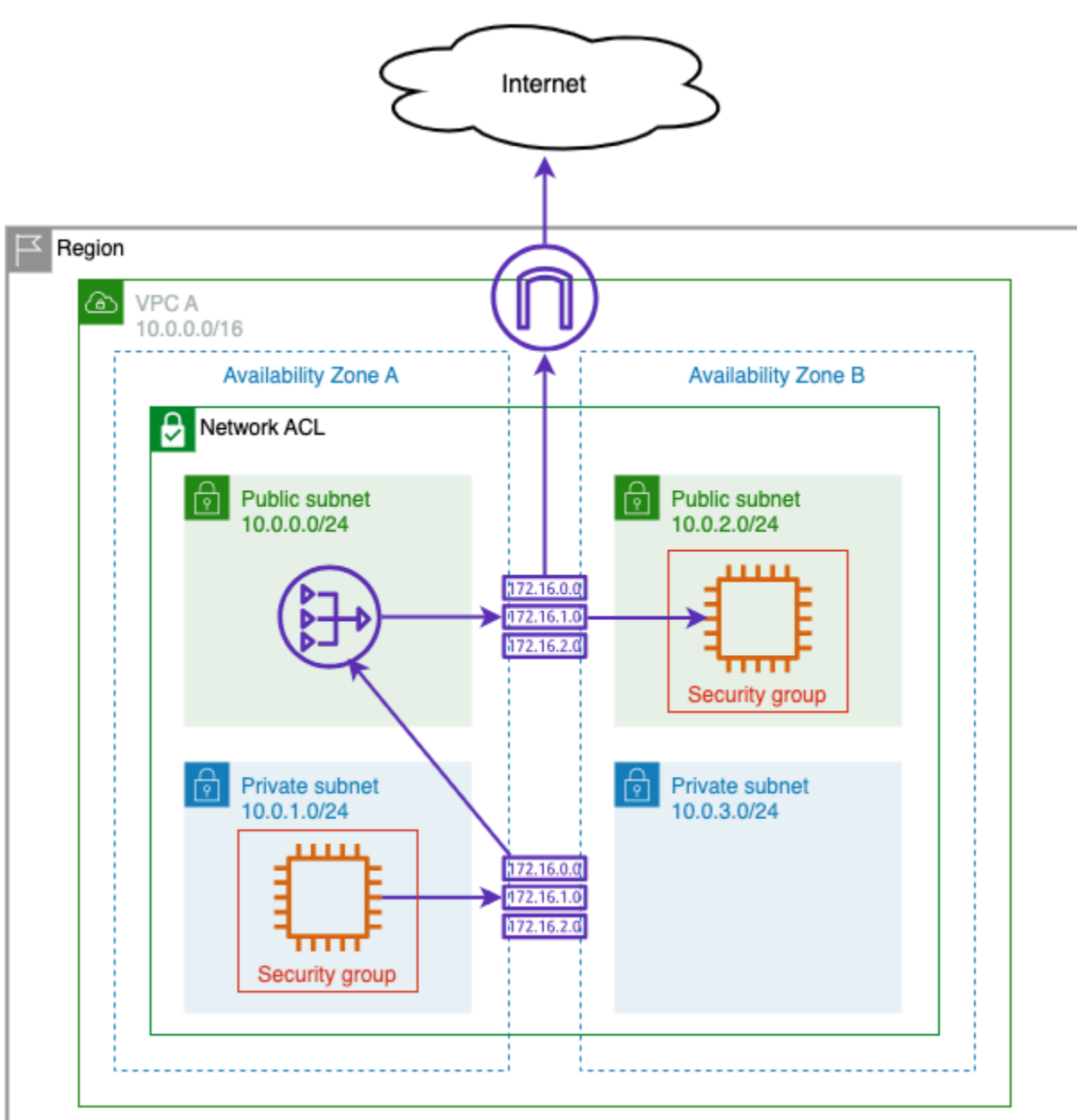
```
ping 10.0.2.100 -c 5
ping example.com -c 5
```

- You should receive responses both from the public EC2 instance and example.com

```
sh-4.2$ ping 10.0.2.100 -c 5
PING 10.0.2.100 (10.0.2.100) 56(84) bytes of data.
 64 bytes from 10.0.2.100: icmp_seq=1 ttl=255 time=0.763 ms
 64 bytes from 10.0.2.100: icmp_seq=2 ttl=255 time=1.60 ms
 64 bytes from 10.0.2.100: icmp_seq=3 ttl=255 time=0.761 ms
 64 bytes from 10.0.2.100: icmp_seq=4 ttl=255 time=0.896 ms
 64 bytes from 10.0.2.100: icmp_seq=5 ttl=255 time=0.792 ms
--- 10.0.2.100 ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4041ms
 rtt min/avg/max/mdev = 0.761/0.964/1.609/0.326 ms
sh-4.2$ ping example.com -c 5
PING example.com (93.184.216.34) 56(84) bytes of data.
 64 bytes from 93.184.216.34 (93.184.216.34): icmp_seq=1 ttl=45 time=28.4 ms
 64 bytes from 93.184.216.34 (93.184.216.34): icmp_seq=2 ttl=45 time=23.6 ms
 64 bytes from 93.184.216.34 (93.184.216.34): icmp_seq=3 ttl=45 time=23.6 ms
 64 bytes from 93.184.216.34 (93.184.216.34): icmp_seq=4 ttl=45 time=23.7 ms
 64 bytes from 93.184.216.34 (93.184.216.34): icmp_seq=5 ttl=45 time=23.6 ms
--- example.com ping statistics ---
 5 packets transmitted, 5 received, 0% packet loss, time 4007ms
 rtt min/avg/max/mdev = 23.640/24.639/28.431/1.906 ms
sh-4.2$
```

If the ping to 10.0.2.100 is not successful, please ensure you correctly configured the IP address in the [EC2 Instances](#) section

Congratulations you have now confirmed outbound connectivity from both instances



Compare Gateway and Interface Endpoint Approaches

Interface Endpoint DNS

- Type the following to check the DNS for the KMS service from the VPC A instance:

```
dig kms.us-east-1.amazonaws.com
```

- The response should point to two local IP addresses within the VPC A private subnet CIDR blocks of 10.0.1.0/24 and 10.0.3.0/24

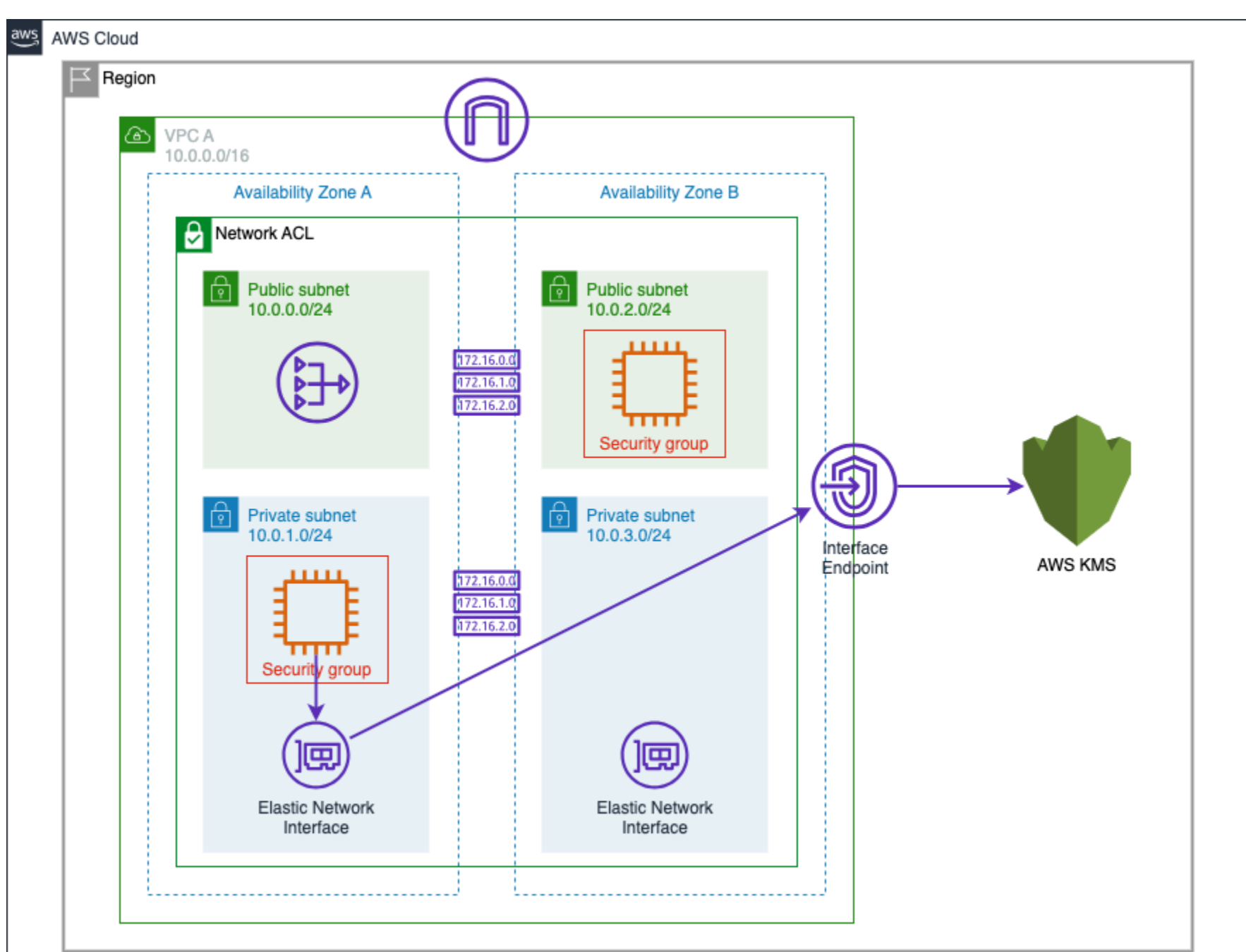
```
sh-4.2$ dig kms.us-east-1.amazonaws.com
;<>> Dig 9.11.4-P2-RedHat-9.11.4-P2.amzn2.5.2 <>> kms.us-east-1.amazonaws.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25901
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 4096
;; QUESTION SECTION:
;kms.us-east-1.amazonaws.com. IN A

;; ANSWER SECTION:
kms.us-east-1.amazonaws.com. 60 IN A 10.0.3.47
kms.us-east-1.amazonaws.com. 60 IN A 10.0.1.111

;; Query time: 2 msec
;; SERVER: 10.0.0.2#53(10.0.0.2)
;; WHEN: Tue Sep 21 07:20:55 UTC 2021
;; MSG SIZE rcvd: 88
```

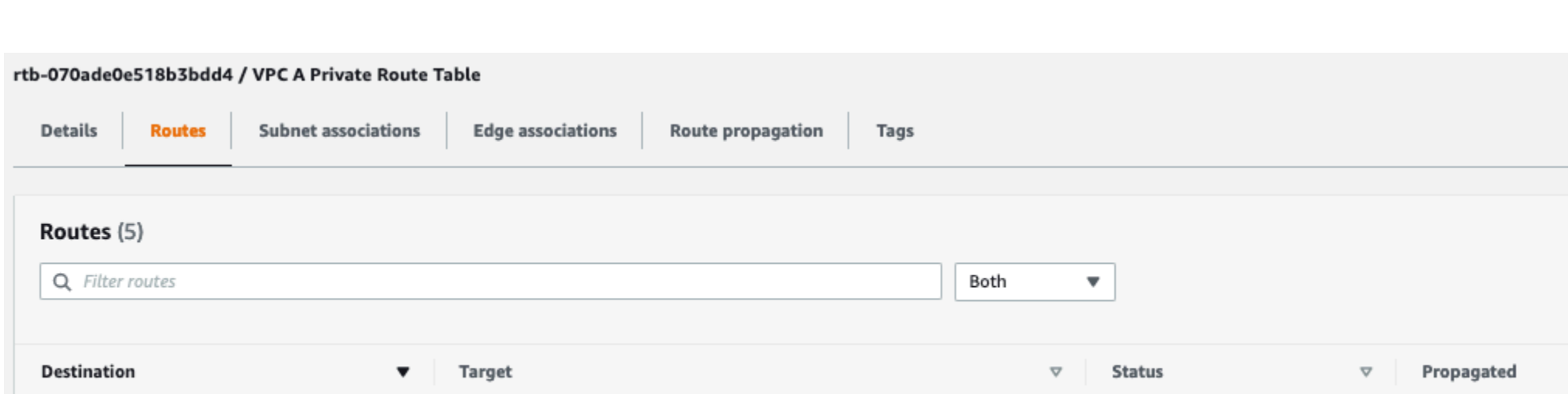
Because the VPC DNS is returning the IP addresses for the Elastic Network Interfaces placed into the private subnets by the Interface Endpoint rather than the public IP address for KMS, the traffic to KMS from VPC A will be routed to the VPC Endpoint rather than traversing the internet to reach KMS.



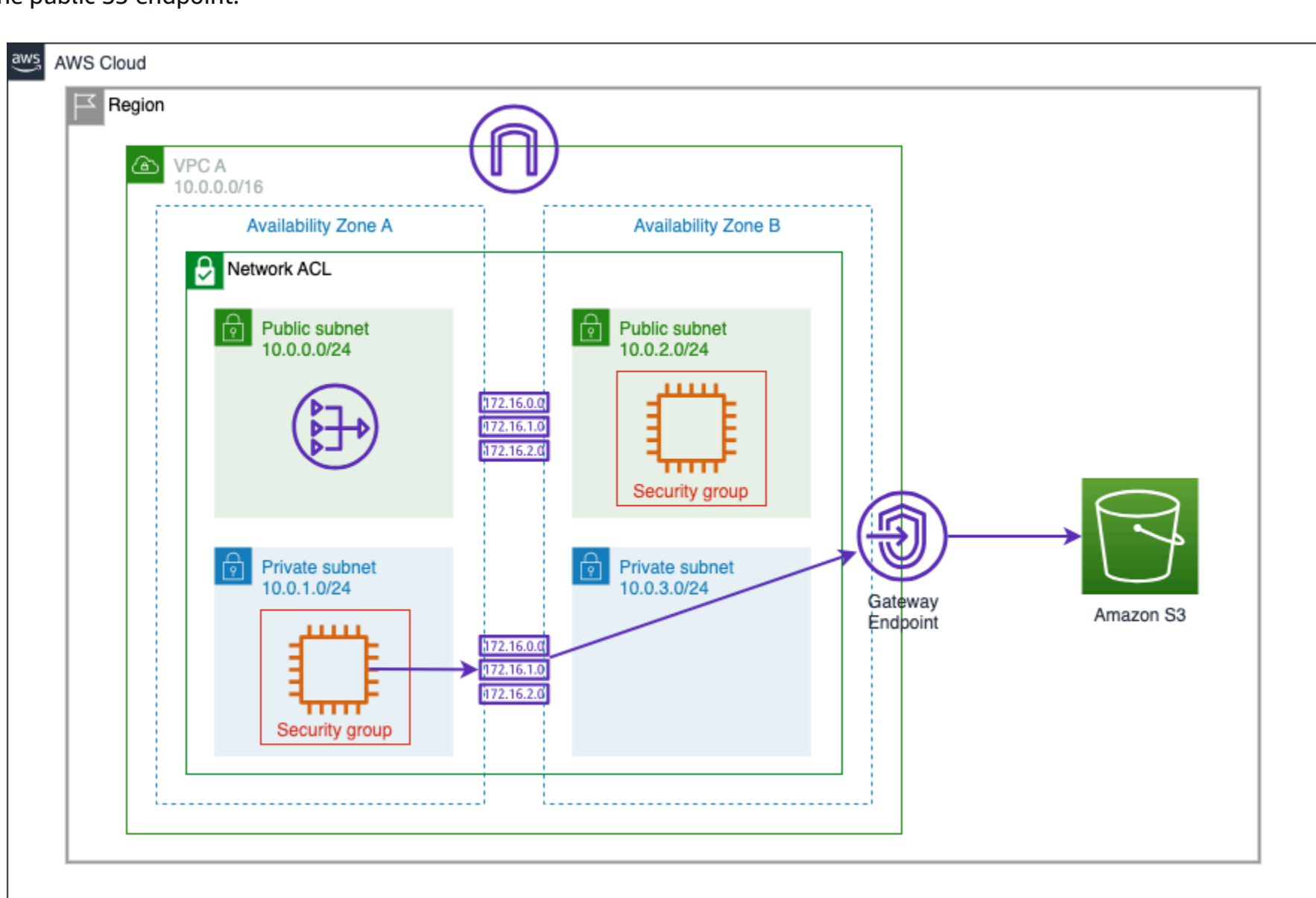
- Terminate the Session Manager session.

Gateway Endpoint Routing

- Navigate to the VPC console, and select [Route Tables](#)
- Select the **check box** next to VPC A Private Route Table and scroll down to the **Routes** tab. It should show an entry pointing to the VPC Endpoint ID for destinations in the prefix list.



Because the S3 prefix list is a more specific route to S3 than the default route to the internet the traffic to S3 from VPC A will be routed to the VPC Endpoint rather than traversing the internet to the public S3 endpoint.



We have created VPC Endpoints for S3 and KMS and confirmed that the local route table pointed to the VPC endpoint for S3 whereas the local DNS entry for KMS now pointed